Estimating Investor Demand Elasticity with Endogenous Firm Responses*

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Abstract

Inelastic investor demand for stocks implies that demand shocks can move prices substantially, but standard estimates treat firms' equity supply as fixed, ignoring that firms respond to that same price pressure by issuing or repurchasing equity. Using mutual fund flow shocks, I show that a one-standard-deviation flow shock raises stock prices by 1.9% on impact and equity issuance by 0.2% of assets, with firms also raising investment and paying down debt. I build a dynamic model of strategic interaction between firms and investors to interpret these facts: firm issuance improves fundamental value, so observed market value blends price pressure with the fundamental-value response it triggers, violating the exclusion restriction standard estimation relies on. Correcting for this bias raises the estimated elasticity from a naive 1.3 to 1.6, since fundamental-value improvement accounts for 17% of the initial price response. This corrected elasticity is not a single number: it is source-, size-, and state-dependent, with investors far more responsive to price pressure than to fundamental value, and firms' and investors' self-limiting responses causing marginal price effects to diminish as shocks grow larger. Together, the corrected elasticity and its self-limiting nature imply that inelastic investor demand is not a first-order source of capital misallocation: the scale and absorption of investor demand shocks move aggregate TFP an order of magnitude less than a comparable change in real investment frictions.

Keywords: Demand Elasticity, Structural Estimation, Equity Issuance, Market Timing, Mutual Fund Flows, Capital Misallocation
JEL classification: G32, G23, G12

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Introduction

The demand-based asset pricing literature (Koijen and Yogo, 2019) documents that investor demand for stocks is inelastic, implying that demand shocks can drive substantial price fluctuations. Meanwhile, on the supply side of the equity market, firms actively time the market when issuing and repurchasing equity, strategically responding to demand-driven valuation changes (Bolton et al., 2013). When temporary demand pushes a firm's *market value* above what its fundamentals justify, the resulting gap, *price pressure*, makes issuing equity cheap. Additional supply absorbs some of this pressure mechanically. Firms that channel the proceeds into investment or debt paydown go further: they improve their capacity to generate future cash flows, a lasting change in *fundamental value*. Standard approaches to measuring demand elasticity treat a firm's equity supply as fixed, abstracting from this channel entirely.

Incorporating this equity-supply channel raises two questions that share a common root. First, an identification question on the demand side: how should demand elasticity be measured when observed market value blends price pressure with the fundamental-value response it triggers? Second, a real-effects question on the supply side: does this uncorrected wedge push firms toward over- or under-investment, and how much capital misallocation does that imply in aggregate? This paper develops and estimates a dynamic structural model of firm-investor interaction: separating price pressure from fundamental value is simultaneously the fix for the identification question and the mechanism behind the real-effects question.

Correcting for firm issuance answers both. The resulting elasticity is higher than a standard approach would recover: 1.6 rather than 1.3. More importantly, how investors react to price changes cannot be summarized by a single number: it depends on the shock's source, its size, and the firm's state. That heterogeneity is what ultimately keeps inelastic demand from being a major source of capital misallocation.

Mutual fund flow shocks provide empirical evidence of this response. Using a shift-share design built on these shocks, I estimate impulse response functions (Jordà, 2005) following Dou et al. (2022) and Chaudhary et al. (2022). A one-standard-deviation flow shock raises a firm's stock price by 1.9% on impact, with only partial reversal, and firms respond by issuing equity (0.2% of assets), raising investment, and paying down debt. These responses are strongest among firms with high mutual fund ownership and high growth opportunities. A fixed-supply approach to measuring demand elasticity would miss this issuance response entirely.

To interpret this issuance response, I build a model with three types of agents: firms, mutual funds, and residual investors. Firms maximize fundamental value through investment and financing, facing equity issuance costs that fall when the firm is overvalued and repurchase costs that fall when it is undervalued (Hayashi, 1982; Zhang, 2005; Belo et al., 2019). Mutual funds receive the exogenous flow shocks and adjust portfolios subject to benchmarking constraints that prevent them from immediately absorbing flows. A residual pool of other investors absorbs what mutual funds do not, subject to a trading cost that rises with the size of their trade, generating a downward-sloping demand curve with respect to price pressure; market clearing across the three agent types then pins down the market value, and with it the resulting price pressure.

This model implies four properties governing the estimation and interpretation of demand elasticity. A transparent two-period version establishes the first three: firm issuance violates the exclusion restriction, and elasticity is both source- and size-dependent. Because this two-period setting is highly stylized, credibly matching the data requires extending it into a full dynamic model with additional structure. I estimate that model via indirect inference, using firms' and investors' optimization conditions as identifying moments. The richer dynamics also deliver a fourth: elasticity is state-dependent as well, shaping both how firms use this financing and how investors respond to demand shocks.

The natural elasticity of interest is investors' response to demand-driven price pressure alone, holding a firm's fundamentals fixed, exactly the ceteris paribus condition the exclusion restriction is meant to guarantee. Standard estimation cannot isolate it: even when the underlying demand shock is exogenous, firm issuance responds to price pressure itself, so market value, the only observable, blends price pressure with the fundamental-value response it triggers. That fundamental-value response accounts for 17% of the initial price response to a flow shock. Attributing that portion to price pressure overstates how far price pressure moves for a given shock, biasing the recovered elasticity down. Instrumenting observed market value changes with flow shocks, the standard approach, recovers a coefficient of 1.3, versus the true elasticity of 1.6 once the two are correctly separated.

This corrected elasticity is still only an average: it is also source-, size-, and state-dependent. That heterogeneity matters for what inelastic demand really implies for price dynamics and for the counterfactuals. Consider source first: the same pressure-fundamental distinction behind the bias means investors respond to the two very differently. Residual investors respond aggressively to price pressure, with an elasticity of 1.6. A price rise driven by transitory demand implies the price should revert, so investors sell into it. Mutual funds respond to fundamental value only modestly. When a firm experiences a positive productivity shock and organically expands, its expected return and volatility both fall, and funds trim their positions a little to balance that trade-off, producing an elasticity of just 0.7, below their sensitivity to observable market value.

Size dependence reflects self-limiting strategic responses: as a shock grows, its price impact shrinks, implying a higher effective elasticity at the margin. As a firm issues more equity in response to price pressure, the additional supply itself absorbs some of that pressure, so each further unit of issuance is progressively less rewarding; investors, anticipating a sharper reversal as price pressure grows, also push back harder. Together, these forces mean that even holding residual investors' underlying structural elasticity fixed, price pressure and equity issuance do not scale proportionally with shock size. This self-limiting property will matter directly for how much real misallocation inelastic demand can generate. This pattern also appears in a non-targeted empirical check: the per-unit price response to small shocks is around 60% larger than to shocks of average size.

Firm state shapes firm actions, which in turn determine the return-volatility profile investors face and, through it, their effective elasticity to price movements. Two untargeted moments, one tied to growth opportunities and one to leverage, confirm the model matches the data. Growth opportunities lower that elasticity on net, since issuance both eases price pressure and directly improves fundamental value, producing the stronger return and issuance response observed for high-Tobin's-Q firms. Leverage splits into two forces. Debt paydown similarly lifts fundamental value, so high-leverage firms show a stronger return response on average. But leverage also makes investors wary of price increases, and issuing further only erodes the elevated price pressure faster, leaving equity issuance and the fundamental-value gains it would support weaker. This composition resolves an ostensibly puzzling pairing: a stronger return response alongside weaker equity issuance for high-leverage firms.

Together, the corrected elasticity and its source-, size-, and state-dependence answer the identification question; what remains is their implication for real effects. An elasticity of 1.6, even corrected upward, is still far from the frictionless benchmark of infinitely elastic demand. One might expect that imperfect elasticity allows prevalent demand shocks to translate into wide swings in price pressure, giving firms ample incentive to over- or under-invest and, in aggregate, causing capital misallocation. The actual effect is far smaller, since the corrected elasticity is higher than a standard approach would recover and firms' and investors' self-limiting responses cap how far that translates as shocks grow. Quantitatively, financial-side counterfactuals (higher fund flow shock volatility, a smaller mutual fund sector, a less elastic residual-investor sector) move aggregate TFP by less than 0.03% in either direction, an order of magnitude smaller than the 0.14% gain available from a comparable reduction in real capital adjustment costs—and substantially more moderate than the misallocation Choi et al. (2021) find in a related setting.

This paper contributes to several literatures. First, it speaks to the demand-based asset pricing literature establishing that investor demand for individual stocks is inelastic (Koijen and Yogo, 2019). The literature since has studied interactions among market participants, with Haddad et al. (2021) finding weaker-than-expected strategic reactions among institutional investors. This paper contributes by incorporating firms' supply-side responses, linking inelastic demand to real distortions through endogenous equity issuance.

A closely related paper is Choi et al. (2021), who combine a dynamic firm model with the logit demand system of Koijen and Yogo (2019), and find large economy-wide misallocation due to investor demand fluctuations. My approach differs in two ways. Instead of taking the logit demand system as given, I explicitly model investors' optimization problem and the resulting equilibrium price interactions. Doing so lets me distinguish elasticity with respect to fundamental value from elasticity with respect to price pressure, the parameter directly relevant for misallocation counterfactuals, and it generates self-limiting responses, in which price impact diminishes as a shock grows rather than staying constant as a logit specification would imply. And rather than attributing all unobserved heterogeneity to demand shocks, I focus specifically on mutual fund flow shocks as the source of demand shifts, enabling cleaner identification and a more precise quantitative assessment. Together, these differences reveal substantially more moderate misallocation.

A growing body of contemporaneous work raises identification concerns specific to demand system estimation. Fuchs et al. (2023), Davis et al. (2022), and Haddad et al. (2025) each identify theoretical and empirical pitfalls arising from cross-asset portfolio interactions on the demand side. This paper addresses a complementary source of bias: endogenous equity supply by firms, which operates along an orthogonal dimension but is equally consequential for recovering structural elasticities. He et al. (2025) studies general-equilibrium return dynamics induced by demand shocks, focusing on endogenous investor rebalancing rather than firm real and financing decisions. van Binsbergen et al. (2025) shows that the intertemporal return implications of current price changes are essential for accurate elasticity measurement;

this paper reaches a similar conclusion through a structural model of forward-looking equity issuance. Chaudhry and Li (2025) documents that price multipliers are diminishing in shock size, consistent with the non-linearity that arises in this paper as strategic equity issuance becomes self-limiting at the margin.

Second, this paper connects to the literature on time-varying financing costs, which has primarily focused on firm-side responses while taking the financing cost variation as given. Eisfeldt and Muir (2014) and Begenau and Salomao (2019) study aggregate financing patterns, while Belo et al. (2019) examine the asset pricing implications. Warusawitharana and Whited (2016) make a seminal contribution by using structural models to uncover hard-to-observe misvaluations and their implications on equity issuance. They emphasize misvaluation as a driver of financing cost variability. Empirically, Ma (2019) document how nonfinancial corporations act as cross-market arbitrageurs in their own securities. This paper contributes by modeling the demand side of equity markets, showing that investor demand shocks are a key source of the financing cost variation that firms respond to.

Third, this paper provides a structural interpretation of the empirical literature on price impacts of fund flows. Coval and Stafford (2007), Lou (2010) and Dou et al. (2022) study the asset pricing implications of mutual fund flows. Khan et al. (2012), Edmans et al. (2012) and Lou and Wang (2018) document the effects of flow-induced tradings on firm fundamentals. Chinco and Sammon (2024) and Sammon and Shim (2024) identify passive funds and find that firms' supply is key in clearing passive demand. A related literature examine sentiment-based explanations, including Baker and Wurgler (2003), Frazzini and Lamont (2008), Polk and Sapienza (2009), and Chiu and Kini (2014).

Finally, a recent literature combines supply and demand in segmented bond markets (Vayanos and Vila, 2021; Siani, 2022; Coppola, 2025). The equity market features a unique direct feedback loop: firm issuance decisions directly affect the fundamental value of the shares being traded.

The remainder of the paper is organized as follows. Section 1 introduces data, presents the empirical results, and conducts heterogeneity analyses to provide targets and initial guidance for the model. Section 2 presents a simple two-period model to illustrate the key mechanisms. Section 3 extends the model to a dynamic setting and defines the equilibrium. Section 4 discusses the calibration and estimation results. Section 5 analyzes the policy functions and key mechanisms. Section 6 conducts counterfactual experiments. Finally, Section 7 concludes the paper.

1 Data and Empirical Findings

This section presents empirical findings that both motivate and serve as estimation targets for the model. I document the impact of mutual fund flow shocks on stock prices and equity issuance. The impulse response analysis reveals demand-side frictions and highlights how firms endogenously respond to price movements, underscoring the importance of incorporating firm actions in demand elasticity estimation.

1.1 Data Sources

Mutual fund holdings data are from CRSP Survivor-Bias-Free US Mutual Fund Database. I focus on mutual funds with domestic equity holdings (CRSP style code starting with ”ED” or ”M”). Since holdings data are only available at quarterly frequency in earlier years and are more consistent at quarterly frequency in later years, I aggregate returns and total net assets (TNA) to quarterly level. Following Elton et al. (2001), I require the lagged TNA to be larger than \$15 million. The final sample is from 2003Q1 to 2021Q4. On average, there are 3966 distinct mutual fund portfolios per quarter, a number that grows over time.

Firm characteristics are from CRSP/Compustat Merged Database. Section A.1 provides a detailed description of variable construction. Following Eisfeldt and Muir (2014), Net Equity Issuance (NEI) is defined as Sale of Common and Preferred Stock net of Repurchase of Common and Preferred Stock and Dividends[1].

1.2 Mutual Fund Flow Shocks

Following Dou et al. (2022), for each mutual fund m in period t, flows are calculated by

$$\text{Flow}_{m,t} = \frac{\text{TNA}_{m,t} - \text{TNA}_{m,t-1} \times (1 + R_{m,t})}{\text{TNA}_{m,t-1}}$$

To extract flow shocks $\hat{\eta}_{m,t}$, for each T-quarter rolling window (here $T = 16$), I regress flows on fund and portfolio characteristics $\mathbf{X}_{m,t}$:

$$\begin{aligned}
\text{Flow}_{m,t} &= \mathbf{X}_{m,t}\boldsymbol{\beta}_\tau + \eta_{m,t} \quad \text{for } t = \tau + 1, \tau + 2, \ldots, \tau + T - 1, \\
\hat{\eta}_{m,\tau+T} &= \text{Flow}_{m,\tau+T} - \mathbf{X}_{m,\tau+T}\hat{\boldsymbol{\beta}}_\tau
\end{aligned}$$

[1] Compustat item `sstk - prstkc - dv`

The fund characteristics include lagged flow, fund excess return relative to the market return, and the portfolio characteristics are value-weighted characteristics in the Fama-French five-factor model (i.e. log market equity, book-to-market ratio, profitability, investment, and market beta) (Fama and French, 1993). The portfolio characteristics have significant predictive power and are time-varying as is shown in Figure A.1.

Then I calculate the idiosyncratic component of the flow shocks $f_{m,t}$ (with a slight abuse of notation) by regressing $\hat{\eta}_{m,t}$ on time and fund fixed effects to remove potential common factors

$$\hat{\eta}_{m,t} = \alpha_m + \delta_t + f_{m,t}. \tag{1}$$

Let $S_{i,m,t}$ denote the the number of shares of firm i held by fund m, firm-level shock $f_{i,t}$ is calculated by

$$f_{i,t} = \frac{\sum_m S_{i,m,t-1} f_{m,t}}{\sum_m S_{i,m,t-1}}. \tag{2}$$

After aggregating flow shocks to firm level, I standardize flow shocks for easier comparison. Because shocks are constructed from one-step-ahead predictions using a $T = 16$-quarter rolling window starting in 2003Q1, the final firm-level sample spans 2007Q1 to 2021Q4.

1.3 Firm Responses

Table 1 presents the summary statistics of the firm panel. Mutual funds hold 24.6% of firm equity on average, with large variations across firms and time. Net equity issuance is positive on average (0.94% of total assets) and exhibits large heterogeneity, with firms engaging in substantial issuances and repurchases. As a validation check, I construct an alternative equity measure using book equity changes (total assets net of retained earnings and total liabilities), which exhibits patterns similar to the baseline cash flow-based NEI measure.

Table 1 Here

I estimate the following local projection in the spirit of Jordà (2005)

$$\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-1} + \varepsilon_{m,t+h} \text{ for } h = 0, 1, \ldots, 4, \tag{3}$$

where control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics: log market equity, log book-to-market ratio, cash scaled by asset, and investment. β_h can be interpreted as the $h-$quarter ahead impulse response from the mutual fund flow shocks. The coefficients are plotted in Figure 1. The shaded area indicates 90% pointwise confidence bands using standard errors clustered at firm level.

Figure 1 Here

Panel A shows a one-standard deviation flow shock leads to a 1.9% stock price increase. This price impact suggests constraints on both mutual funds and other investors. When an unconstrained mutual fund receives unexpected inflows, it does not need to scale up its holdings, avoiding positive price pressure. Similarly, if other investors were unconstrained, they should be able to absorb the excess demand from unexpected mutual fund flow shocks. Instead of an immediate price correction, the price reversal in subsequent quarters is much smaller than the initial impact. Section A.3 reports the corresponding increase in mutual fund ownership underlying this price pressure.

Panel B demonstrates that firms react to these price impacts by issuing equity. The size of NEI (a cash flow measure) is 0.2% of total assets, which is corroborated by the growth of total assets (a balance sheet measure, Table 2 Column (5)). This firm response highlights the identification challenge. Even if the initial flow shocks are plausibly exogenous, firms' optimal financing responses generate endogenous changes in fundamental value. Since equity issuance can fund investment or reduce leverage, thereby increasing future cash flows or reducing risks, part of the persistent price increase may reflect improved fundamentals rather than pure price pressure. This complicates the interpretation of demand elasticities.

Table 2 presents the response on impact ($h = 0$ in equation (3)) on additional firm-level variables. Columns (1) and (2) provide further validation for the baseline results on equity issuance. In Column (1), book equity increases by about 0.1%. In Column (2), when focusing on Sale of Common Stock and apply the 2% market equity filter following McKeon (2015), the estimate (0.16%) is slightly smaller than the baseline specification (0.2%) since the share repurchases are excluded. As expected from equity issuance, cash stock goes up in Column (3), while physical capital rises by 0.2% (Column (4)), indicating investment in productive assets. Column (5) shows that total assets, an alternative growth measure, rise by 0.2% as well, corroborating the increase in physical capital. Column (6) shows that leverage decreases by 0.03 percentage points, reducing risks. Moreover, since the leverage decline is smaller than the asset increase, firms are increasing debt financing, possibly with the help of increased collateral capacity.

Table 2 Here

1.4 Heterogeneity Analysis

1.4.1 Positive and Negative Flow Shocks

While NEI is positive on average, Table 1 shows that there is a significant portion of observations with negative NEI, reflecting net repurchases. Do firms respond to flow shocks differently when faced with positive and negative price pressures? To test for such asymmetry, I interact $f_{i,t}$ with two dummy variables, each indicating whether flow shocks are positive or negative.

Table 3 shows the response on impact ($h = 0$ in equation (3)) for positive and negative flow shocks. Column (2) shows that the price impacts are roughly symmetric. However, share repurchases following negative flow shocks are weaker than share issuances following positive shocks. This is consistent with the fact that share repurchases, as a form of payout, are generally smaller than share issuances. Despite the asymmetry, the results demonstrate that firm managers actively respond to price pressures in both directions, exploiting both temporary overvaluations through issuance and undervaluations through repurchases.

Table 3 Here

1.4.2 Active and Passive Funds

Based on fund classification information from WRDS, I separately calculate flow shocks for active and passive funds, with detailed construction and properties described in Section A.1.1. On average, passive and active funds own 9% and 15% of firm equity, respectively. After removing time fixed effects, the correlation between the two shocks is only 0.07. This low correlation provides supporting evidence for the exogeneity of flow shocks, as it suggests that fund flows are not common responses to firm fundamentals.

Figure 2 presents impulse response functions of firm returns to active and passive mutual fund flow shocks ($f_{i,t}^T$ for $T \in \{\text{Active}, \text{Passive}\}$). Despite similar standard deviations (0.362 versus 0.364), active and passive flow shocks generate different return responses. Passive fund flow shocks lead to a 2.0% contemporaneous return jump, nearly double the 1.1% response to active fund flow shocks. The subsequent reversals follow a similar pattern: passive fund flow shocks lead to a -0.5% reversal compared to a more moderate -0.25% reversal for active fund flow shocks.

Figure 2 Here

Passive funds typically operate under strict mandates, so unexpected AUM increases mechanically translate into heightened demand for constituent stocks. However, active funds should theoretically be able to avoid exerting positive price pressures on their holdings. Two factors likely explain why active funds still generate significant price pressures. First, active funds face portfolio constraints that limit their trading flexibility such as dividend reinvestment requirements (Schmickler and Tremacoldi-Rossi, 2022), benchmark tracking constraints (Kashyap et al., 2021), and client redemption pressures (Edmans et al., 2012). The weaker return responses relative to passive funds likely reflect these less stringent but still effective constraints. Second, as documented by Chinco and Sammon (2024), some nominally active funds are in fact index-tracking. Consequently, the impulse responses potentially reflect a mixture of truly active and passive investment strategies.

1.4.3 Firm Characteristics

To examine whether the baseline effects vary across firm types and provide suggestive evidence on mechanisms that are important for the model, I estimate the baseline specification (3) across subsamples based on key firm characteristics. Within each Fama-French 12 industry, I split the sample at the median of three lagged variables: mutual fund ownership, Tobin's Q, and leverage. Table 4 reports the return and NEI responses on impact ($h = 0$).

Table 4 Here

Panel A shows that firms with higher mutual fund ownership mechanically exhibit stronger responses in both returns and NEI. In Panel B, firms with higher growth potential (high Tobin's Q) demonstrate stronger responses across both dimensions. Return responses are about 50% stronger for the high-Q firms. Low-Q firms show essentially no reaction to price pressures through equity issuance, statistically indistinguishable from zero, consistent with their reduced need for external financing to fund expansion opportunities. While part of this pattern reflects the ownership channel as high-Q firms attract somewhat more mutual fund investment, this modest difference in ownership cannot fully account for the observed differences.

Panel C presents the results for subsamples partitioned by leverage. High-leverage firms have stronger return responses but weaker equity issuance responses than low-leverage firms, a pattern that at first appears puzzling, since the group with the larger price response reacts *less* through issuance. Mean mutual fund ownership is similar across the two groups, so this pattern is not primarily ownership-driven as in Panels A and B. Instead, as Section 5 shows, it reflects a composition of two offsetting forces that depend on firms' growth opportunities: low leverage provides financing flexibility that is valuable mainly for firms with investment opportunities to fund, while for firms with fewer such opportunities, high leverage instead creates room to benefit from paying down debt at favorable prices. Averaging across firms with different growth opportunities produces the pattern in this panel.

1.5 Additional Tests

The shift-share instrument is valid if it captures exogenous variation in investor demand, uncorrelated with changes in firm fundamentals. Validity can be established via exogenous shares (Goldsmith-Pinkham et al., 2020) or exogenous shifts (Borusyak et al., 2018, 2025). Section A.2 provides supporting evidence for both, and results are stable across all tests. As a further check, Section 5 shows that stock return and equity issuance responses estimated on small fund flow shocks alone are consistent with the model's self-limiting mechanism.

Under the share-based design, the concern is that lagged ownership shares reflect anticipatory allocation toward firms expected to perform well. Including characteristic-by-quarter fixed effects (industry, size, book-to-market, and profitability) absorbs systematic portfolio tilts along those dimensions. Reconstructing shocks using shares lagged four quarters addresses near-term firm-specific repositioning, with mild attenuation consistent with classical measurement error.

Under the shift-based design, the concern is that fund flows are correlated with anticipated firm fundamentals. Industry-by-quarter fixed effects address the concern that investors direct flows into industry-themed funds in anticipation of sectoral performance. Identification then comes from within-industry variation. Removing the largest firms within each industry addresses the possibility that flows are driven by the salient performance of superstar firms. There is also no pre-trend in firm fundamentals prior to flow shocks.

I also examine the sensitivity of the baseline results to alternative NEI definitions. First, excluding dividend payments from the baseline measure yields virtually identical findings, as expected given that dividends are less volatile than equity issuances and repurchases. Second, I construct gross equity issuance using the 2% market equity cutoff of McKeon (2015) and find similar results. I do not adopt this as the baseline, however, because the cutoff cannot be applied to equity repurchases, and other forms of stock issuance also represent costly equity financing[2].

2 A Two-Period Model

To illustrate the mechanisms in a simple setting, I present a two-period model with minimal ingredients and near closed-form solutions. There are three types of agents: a firm making investment decisions, a mutual fund and a continuum of residual investors trading the firm's equity. Guided by empirical evidence, the model's core mechanism focuses on how exogenous fund flow shocks affect asset prices when trading constraints limit residual investors' arbitrage capacity. These price movements influence the firm's equity issuance costs, leading to changes in fundamental values that feed back into investor demand.

The model delivers three important properties. First, the comparative statics highlight the identification challenge: even with perfectly exogenous demand shocks, endogenous firm responses bias estimates of demand elasticities. Second, efforts by both investors and the firm to trade against price pressure attenuate price responses, limiting the real effects of demand shocks. Finally, the notion of demand elasticity depends on the source of price changes, as different sources have different implications for expected returns.

2.1 Model Setup

There is a firm, a mutual fund, and a continuum of residual investors. The timeline is as follows. In the first period, the firm first chooses investment, then the mutual fund decides its portfolio allocation, and finally the residual investors trade to clear the market. Absent frictions, residual investors would trade to ensure the firm is priced at its fundamental value P. However, when the mutual fund experiences a demand shock, the residual investors' limited trading capacity prevents full price correction, causing the market value $\tilde{P}$ to deviate from P. The ratio $\Lambda = \tilde{P}/P$ measures the price pressure and determines the issuance cost for the firm and returns for the mutual fund. In the second period, the firm produces output and distributes all profits.

The firm. The firm has initial capital K and therefore production K^{α}, where $\alpha \in (0, 1)$. With full depreciation, investment I becomes next-period capital K' as in (5). Net income

[2]For more discussions, see Huddart (1994); Fama and French (2005); Billett et al. (2019); Belo et al. (2019); Sammon and Shim (2024), among others.

E equals production minus investment, given by (6). When $E < 0$, the firm issues equity to cover the shortfall, incurring cost Ψ in (7). The cost decreases in price pressure Λ, as higher prices allow the firm to raise the same funds by issuing fewer shares. The firm maximizes its value V in (4), the sum of current and discounted future payouts. Payout in the first period equals net income minus any issuance cost. The present value of future payout is the expected value of next-period production $A'K'^{\alpha}$ discounted by the stochastic discount factor (SDF) M', where A' is a productivity shock with $\log A' \sim \mathcal{N}(0, \sigma_a^2)$.

$$\max_{I} \quad V = E - \mathbb{1}_{\{E<0\}}\Psi + \mathbb{E}[M'A'K'^{\alpha}] \tag{4}$$

$$\text{s.t.} \quad K' = I \tag{5}$$

$$E = K^{\alpha} - I \tag{6}$$

$$\Psi = \frac{c_h}{2}\left(\frac{E}{\Lambda K}\right)^2 K \tag{7}$$

This setup yields some convenient simplifications. The ex-dividend price

$$P = \mathbb{E}[M'A'I^{\alpha}] = \beta I^{\alpha} \tag{8}$$

depends only on $\beta \equiv \mathbb{E}[M'A']$, not on the SDF or productivity shock separately. Combined with full profit distribution and no trading in the second period, this simplifies the return calculation:

$$R = \frac{V'}{\tilde{P}} = \frac{A'I^{\alpha}}{\Lambda\beta I^{\alpha}} = \frac{A'}{\beta\Lambda} \implies \log R = \log A' - \log\beta - \log\Lambda. \tag{9}$$

Consequently, expected log return $\mu = \mathbb{E}[\log R]$ is independent of investment I, and the log return variance $\sigma^2 = \mathrm{Var}(\log R) = \sigma_a^2$ is independent of both investment I and price pressure Λ.

The mutual fund. The mutual fund chooses its portfolio weight ϕ to maximize its mean-variance preferences over log returns, as in (10), where γ is the risk aversion parameter and r_f is the risk-free rate. The fund also faces a quadratic adjustment cost with parameter c_ϕ, capturing benchmarking incentives tied to a target portfolio weight $\bar{\phi}$.

$$\max_{\phi} \quad \phi(\mu - r_f) - \frac{\gamma}{2}\phi^2\sigma^2 - \frac{c_\phi}{2}(\phi - \bar{\phi})^2 \tag{10}$$

Let Ω denote the fund's size, the share of the firm held by the fund is caculated as $s^{M'} = \phi\Omega/\tilde{P}$.

Residual investors. There is a continuum of one-period living residual investors representing all other shareholders. These investors price firms using the exogenously specified stochastic discount factor and trade accordingly, consistent with the firms' inherited preferences. Absent trading costs, they would trade to ensure that each firm's market value equals its fundamental value. However, trading constraints limit their arbitrage capacity. When mutual funds increase (decrease) their holdings, possibly due to positive (negative) flow shocks, residual investors cannot adjust their positions sufficiently, creating temporary overvaluation (undervaluation).

Residual investors begin with shares s^R and choose new shares $s^{R'}$. Recognizing the firm's fundamental value P and given the market price $\tilde{P}$, they act as fundamental investors aiming to trade toward fundamental value. Their limited trading capacity is captured by a quadratic adjustment cost with parameter c_r (Gârleanu and Pedersen, 2013). They solve:

$$\max_{s^{R'}} \quad s^{R'}(P - \tilde{P}) - \frac{c_r}{2}\left(s^{R'} - s^R\right)^2 \tilde{P} \tag{11}$$

Market Clearing. Market clearing requires that the total demand for shares equals the total supply,

$$s^{M'} + s^{R'} = 1. \tag{12}$$

Equilibrium. An equilibrium consists of investment I^*, mutual fund portfolio weight ϕ^*, residual investor shares $s^{R'*}$, and market price pressure Λ^*, such that

1. Given optimal responses of the mutual fund and residual investors, the firm's investment I^* solves its optimization problem (4)–(7).

2. Given firm investment I^* and optimal responses of residual investors, the mutual fund's portfolio weight ϕ^* maximizes its objective (10).

3. Given firm investment I^* and fund portfolio weight ϕ^*, the residual investors' shares $s^{R'*}$ solve their optimization problem (11).

4. Market clearing (12) holds.

2.2 Optimality Conditions

Following backward induction, I start with the optimality conditions of the residual investors, then the mutual fund, and finally the firm.

Residual investors. The residual investor optimality condition is

$$s^{R'} = s^R + \frac{P - \tilde{P}}{c_r \tilde{P}}. \tag{13}$$

Substituting into the market clearing condition (12) yields price pressure as a function of the mutual fund's portfolio adjustment:

$$\tilde{P} = \frac{1}{1 + c_r(s^M - s^{M'})} P \equiv \Lambda P. \tag{14}$$

When $c_r \to 0$, residual investors have unlimited trading capacity. If the firm is overvalued ($\tilde{P} > P$) residual investors short the firm to eliminate price pressure, driving $\tilde{P}$ toward P. The same mechanism applies when the firm is undervalued.

When mutual fund ownership does not change, $s^{M'} = s^M$, no forces push prices away from fundamental values. Even with positive trading costs, there is no price pressure, $\tilde{P} = P$. In contrast, a positive mutual fund flow shock (larger Ω) increases mutual fund ownership to $s^{M'} > s^M$, as portfolio adjustment costs prevent mutual funds from swiftly adjusting their portfolio weights. With finite trading capacity ($c_r > 0$), residual investors cannot fully trade away this price pressure, causing $\tilde{P} > P$.

The mutual fund. Internalizing how its actions affect price pressure, the mutual fund's optimality condition is

$$\phi = \frac{\mu - r_f + c_\phi \bar{\phi}}{\gamma\sigma^2 + c_\phi + \frac{d \log \Lambda}{d\phi}}. \tag{15}$$

The first two terms in the numerator and the first term in the denominator represent the standard risk-return trade-off. The third term in the numerator and the second term in the denominator capture the effects of portfolio mandates. The mutual fund chooses a weighted average of the mean-variance optimal portfolio and the target portfolio $\bar{\phi}$.

The final term in the denominator reflects the strategic interaction with residual investors. Since $\log \Lambda$ enters return via (9), and Λ depends on the mutual fund's choice via (14), the mutual fund internalizes this feedback. The derivative $d \log \Lambda / d\phi > 0$ (proven in Section B.1) implies that increasing portfolio weight ϕ raises the firm's current valuation, which in turn reduces the marginal benefit of further increasing ϕ.

The firm. The firm's optimality condition is

$$\beta\alpha I^{\alpha-1} = 1 + \mathbb{1}_{\{E<0\}} \frac{c_h}{K} \left[\frac{-E}{\Lambda^2} - \frac{\Lambda \frac{d\Lambda}{dI}}{\Lambda^4} E^2 \right], \tag{16}$$

where the left-hand side is the standard marginal benefit of investment and the right-hand side is the marginal cost. When $E \geq 0$, the marginal cost is 1 as there is no capital adjustment cost. When $E < 0$, the firm raises equity and incurs issuance costs. As with the mutual fund, the final term reflects that the firm internalizes how investment affects price pressure and issuance costs. The derivative $d\Lambda/dI < 0$ (proven in Section B.1) because higher investment requires supplying more shares, which reduces price pressure by absorbing some of the excess demand, and thereby raising issuance costs.

2.3 Model Properties

To make sure that equity issuance is relevant, I focus on parameter values such that the firm issues equity in equilibrium. The comparative statics in Property 1 and Property 2 are derived under the regularity condition that higher-order feedback effects through $\frac{d\Lambda}{dI}E^2$ are small relative to the first-order effects through E. All proofs and the regularity conditions are in Section B.1.

Property 1 (Source of Bias)**.** *Both equilibrium price pressure Λ^* and fundamental value P^* increase in mutual fund size Ω.*

Property 2 (Mitigating Forces)**.** *When fund size Ω increases, the mutual fund optimally reduces its portfolio weight ϕ^*, and the firm optimally increases investment I^*, both partially offsetting the increase in price pressure Λ^*. Formally,*

$$\underbrace{\frac{d\Lambda^*}{d\Omega}}_{>0} = \underbrace{\frac{\partial\Lambda^*}{\partial\Omega}}_{>0} + \underbrace{\frac{\partial\Lambda^*}{\partial\phi^*}}_{>0} \cdot \underbrace{\frac{d\phi^*}{d\Omega}}_{<0} + \underbrace{\frac{\partial\Lambda^*}{\partial I^*}}_{<0} \cdot \underbrace{\frac{dI^*}{d\Omega}}_{>0} .$$

Proof for Property 1 and Property 2 (Sketch). The equilibrium is a system of two unknowns (I, ϕ) and two equations: the firm optimality condition (16) and the mutual fund optimality condition (15), with Λ given by (14). The comparative statics follow from the implicit function theorem. □

Property 1 illustrates the identification challenge. When price pressure rises due to mutual fund inflows, equity issuance costs fall, encouraging higher investment and improving fundamental value. Consequently, the observed price response reflects both exogenous demand pressure and endogenous fundamental improvements.

Property 2 shows the strategic interactions that attenuate price responses. Higher price pressure reduces expected returns for the mutual fund, leading it to reduce its portfolio weight. At the same time, lower issuance costs incentivize the firm to invest more. Both actions absorb part of the excess demand, thereby reducing price pressure.

This property also has important implications for understanding the real effects of demand shocks. With inelastic demand for residual investors, issuance costs for the firm and benchmarking constraints for the mutual fund allow deviations of market prices from fundamental values for small shocks. However, as demand shocks grow larger, the mitigating forces also become stronger, potentially limiting the magnitude of price pressure and consequent real effects.

Property 3 (Context-Dependent Elasticities)**.** *For the residual investors, the elasticity with respect to price pressure Λ is negative, while the elasticity with respect to fundamental value P is zero. For the mutual fund, the elasticity with respect to price pressure Λ is negative, while the elasticity with respect to fundamental value P is positive.*

Proof (Sketch). The elasticities for residual investors follow directly from their optimality condition (13). For the mutual fund, the elasticity with respect to fundamental value P follows from the optimality condition (15). The elasticity with respect to price pressure Λ is derived by considering an exogenous change in $\log \Lambda$. □

Property 3 highlights that the notion of demand elasticity depends on the nature of price changes. The distinction for residual investors is by construction, as they only trade to correct deviations from fundamental value. However, the distinction naturally arises for the mutual fund as well. When the mutual fund expects a sharp correction in price pressure—a significant reduction in returns—it reduces its portfolio weight. In contrast, a rise in fundamental value does not necessarily reduce returns. In particular, higher fundamental value driven by more supply of equity can absorb some of the price pressure, increasing returns. In more general settings where fundamental value changes for other reasons, which then have different implications for expected returns, the elasticity with respect to fundamental value can be positive or negative[3].

Together, these three properties map onto the paper's main results. Property 1 is the source of the identification bias: because equity issuance responds to price pressure, the exclusion restriction underlying standard instrumental-variable estimates is violated, and any resulting elasticity estimate conflates price pressure with fundamental value, a gap the structural estimation in Section 4 quantifies directly. Property 2 previews the size-dependence documented in Section 5 and Section 6: because both the mutual fund and the firm push back harder against larger deviations, price pressure and its real effects are self-limiting. Property 3 previews the source-dependence documented throughout: residual investors' elasticity with respect to fundamental value is zero by construction, while the mutual fund's elasticity with respect to fundamental value, estimated in Section 4, need not be zero once fundamental value can change for reasons beyond the firm's own issuance response.

3 Dynamic Model

This section extends the two-period model to a dynamic setting. The main limitation of the two-period model is that the firm side is highly simplified and cannot quantitatively capture the firm's investment, financing, and return dynamics. Therefore, it is difficult to assess the quantitative importance of fundamental value changes versus price pressure in driving empirical price responses, and the resulting implications for demand elasticity and capital misallocation. The dynamic equilibrium framework provides an environment to identify key parameters that would be difficult to estimate from reduced-form impulse responses alone and to evaluate the quantitative importance of the mechanisms.

3.1 Firms

Firms maximize their fundamental value, defined as the infinite sum of discounted payouts, through optimal investment and financing decisions. The costs of equity issuance and buyback depend on the firm's market valuation. When market value exceeds fundamental value, firms face lower equity issuance costs; conversely, undervaluation makes share buybacks less costly.

[3] In the full model in Section 3 below, larger firms have lower expected returns due to lower risk, implying a negative elasticity with respect to fundamental value.

3.1.1 Technology

Firm i uses physical capital $K_{i,t}$ to produce $Y_{i,t}$:

$$Y_{i,t} = A_t X_{i,t} K_{i,t}^{\alpha}, \tag{17}$$

where A_t is the aggregate productivity and $X_{i,t}$ is the idiosyncratic productivity. The logarithm of aggregate productivity $a_t = \log A_t$ follows an AR(1) process,

$$a_{t+1} = (1-\rho_a)\bar{a} + \rho_a a_t + \epsilon_{a,t+1},\ \epsilon_{a,t+1} \sim \mathcal{N}(0, \sigma_a^2), \tag{18}$$

where ρ_a, $\bar{a}$, and σ_a are the persistence, mean, and conditional volatility of the AR(1) process.

Define $x_{i,t} = \log X_{i,t}$ as the logarithm of idiosyncratic productivity, which follows an AR(1) process:

$$x_{i,t+1} = (1-\rho_x)\bar{x} + \rho_x x_{i,t} + \epsilon_{x,i,t+1}, \tag{19}$$

where ρ_x and $\bar{x}$ are the persistence and mean of the AR(1) process. $\epsilon_{x,i,t+1}$ is an i.i.d. normal shock that follows $\epsilon_{x,i,t+1} \sim \mathcal{N}(0, \sigma_x^2)$, where σ_x is the conditional volatility.

Capital accumulation follows

$$K_{i,t+1} = (1-\delta)K_{i,t} + I_{i,t}, \tag{20}$$

where δ is the depretiation rate, $I_{i,t}$ denotes investment, and $e^{\eta_{i,t+1}}$ is an i.i.d. capital quality shock.

Convex investment adjustment costs $G_{i,t}$ are given by

$$G_{i,t} = \frac{c_k}{2}\left(\frac{I_{i,t}}{K_{i,t}}\right)^2 K_{i,t}, \tag{21}$$

where cost parameter c_k determines the speed of adjustment.

3.1.2 Debt Financing

Each period, firms issue one-period debt L_t, which is to be repaid at $t+1$ at interest rate r_f. Following Hennessy and Whited (2005), collateral constraint is given by

$$L_{i,t+1} \leq \varphi K_{i,t+1}, \tag{22}$$

where $\varphi \in (0,1)$ denotes the borrowing capacity. When $L_t < 0$, the firm saves with interest rate $r_s = r_f - \kappa$, where $\kappa \in (0, r_f)$ denotes the wedge between borrowing and saving rates. Finally, $r_l = r_f \mathbb{1}_{\{L_t>0\}} + r_s \mathbb{1}_{\{L_t \le 0\}}$ denotes the applicable interest rate.

3.1.3 Issuance, Repurchase, and Payout

With corporate taxes τ, the operating profit can be expressed as after-tax sales plus depreciation tax shield, net of investment, investment adjustment costs, and changes in debt financing:

$$E_{i,t} = (1-\tau)Y_{i,t} + \tau\delta K_{i,t} + \tau r_f L_t \mathbb{1}_{\{L_t>0\}} - I_{i,t} - G_{i,t} + L_{i,t+1} - (1+r_l)L_{i,t}. \tag{23}$$

If total operating costs exceed total profits, $E_{i,t} < 0$, firms can raise external equity,

$$H_{i,t} = \max\{0, -E_{i,t}\}, \tag{24}$$

which incurs costs

$$\Psi_{i,t} = c_{h0}K_{i,t} + \frac{c_{h1}}{2}\left(\frac{H_{i,t}}{\Lambda_{i,t}K_{i,t}}\right)^2 K_{i,t}, \tag{25}$$

where c_{h0} and c_{h1} are the fixed and variable issuance cost parameters and $\Lambda_{i,t}$ is the price pressure factor that fluctuates around 1 (an equilibrium outcome defined in Section 3.4). When a firm is overvalued ($\Lambda_{i,t} > 1$), the issuance size $H_{i,t}$ becomes relatively smaller compared to the firm's total market value, making issuance less costly. I adopt a reduced-form specification to capture the empirical relationship in Section 1.4.3, consistent with micro foundations in Baker and Wurgler (2002); Kim and Weisbach (2008); Bolton et al. (2013).

If the operating profit is positive, $E_{i,t} > 0$, following Warusawitharana and Whited (2016), the firm can pay out to shareholders by choosing any combination of dividend $D_{i,t}$ and share repurchase $B_{i,t}$ such that

$$D_{i,t} + B_{i,t} \le E_{i,t}. \tag{26}$$

Dividend is subject to dividend tax τ_D, and buyback $B_{i,t}$ incurs costs

$$\Phi_{i,t} = c_{b0}K_{i,t} + \frac{c_{b1}}{2}\left(\frac{\Lambda_{i,t}B_{i,t}}{K_{i,t}}\right)^2 K_{i,t}, \tag{27}$$

where c_{b0} and c_{b1} are the fixed and variable repurchase cost parameters. Similar to the argument about issuance costs Ψ, when investor demand is weak and the firm is undervalued

($\Lambda_{i,t} < 1$), firms are more incentivized to repurchase stocks at a lower price.

Finally, the payout $O_{i,t}$ is operating profit minus issuance costs when operating profit is negative, or the combination of after-dividend tax dividend and repurchase net of repurchase costs when operating profit is possitive:

$$O_{i,t} = \begin{cases} E_{i,t} - \Psi_{i,t} & \text{if } E_{i,t} \leq 0, \\ (1-\tau_D)D_{i,t} + B_{i,t} - \Phi_{i,t} & \text{if } E_{i,t} > 0. \end{cases} \tag{28}$$

Following Zhang (2005) I specify the stochastic discount factor (SDF) as

$$\log(M_{t+1}) = \log\beta - \gamma_t(a_{t+1} - a_t) - \frac{1}{2}\gamma_t^2\sigma_a^2 \tag{29}$$

$$\gamma_t = \gamma_0 + \gamma_1(a_t - \bar{a}), \tag{30}$$

where $\bar{a}$ is the long-run mean of aggregate productivity. Constant β determines the level of risk-free rate. Constants $\gamma_0 > 0, \gamma_1 < 0$ capture countercyclical price of risk.

Finally, the firm maximizes its continuation value

$$V_{i,t} = \max_{I_{i,t}, K_{i,t+1}} O_{i,t} + \mathbb{E}_t\left[M_{t+1}V_{i,t+1}\right] \equiv O_{i,t} + P_{i,t}, \tag{31}$$

subject to constraints (17) through (30).

3.2 Mutual Funds

There is one one-period living mutual fund for each firm, holding equity positions and earning a fixed fee based on their AUM. Fund AUM is influenced by returns from equity holdings and exogenous fund flow shocks. Idiosyncratic fund flow shocks introduce cross-sectional heterogeneity in fund sizes, providing the variation needed for identification. Position adjustment costs prevent funds from achieving the optimal portfolio allocations.

A mutual fund i manages its inherited AUM $Q_{i,t}$ by allocating its portfolio weights ϕ_{t+1} in firm i and a risk-free asset to gain portfolio returns $R^M_{i,t+1}(\phi_{t+1}) = R_f + \phi_{i,t+1}(R_{i,t+1} - R_f)$.

Let lower case letters denote the logarithm of their upper case counterparts, the size of the mutual fund evolves following

$$\tilde{q}_{i,t+1} = q_{i,t} + r^M_{i,t+1}.$$

The mutual fund manager will then calculate the fixed fee based on $\tilde{q}_{i,t+1}$.

Following the same setup as the SDF [4], up to a constant scaling factor (e.g. a fixed 2% management fee), the fund manager's discounted payoff is then

$$\frac{\beta}{1-\gamma_m}\mathbb{E}_t\left[\tilde{Q}_{i,t+1}^{1-\gamma_m}\right] = \frac{\beta}{1-\gamma_m}\mathbb{E}_t\left[e^{(1-\gamma_m)\tilde{q}_{i,t+1}}\right]. \tag{32}$$

For tractability, I assume that after the returns are realized and the fund managers collect their fees, mutual fund clients will withdraw the returns from the funds and rebalance their portfolios across mutual funds, so that the AUM for each mutual fund available for investment is the sum of an aggregate component and an idiosyncratic component:

$$q_{i,t+1} = \bar{q} + \omega_{i,t+1}, \tag{33}$$

where $\bar{q}$ is the average size of the mutual fund industry and $\omega_{i,t+1}$ is specific to mutual fund i, denoting its deviation from average size.

The idiosyncratic size deviation $\omega_{i,t+1}$ follows an AR(1) process that captures the persistence of each fund's size

$$\omega_{i,t+1} = (1-\rho_\omega)\bar{\omega} + \rho_\omega\omega_t + \epsilon_{\omega,i,t+1}, \tag{34}$$

where $\bar{\omega}$ and ρ_ω are the long run mean and persistence of idiosyncratic fund size. Idiosyncratic fund size shock $\epsilon_{\omega,i,t+1}$ is drawn from an i.i.d. normal distribution with conditional volatility σ_ω.

The amount of proceeds invested to firm i is the portfolio weight for next period $\phi_{i,t+1}$ multiplied by fund size today $Q_{i,t}$. Divide it by the total market value $\tilde{P}_{i,t}$, we get the mutual fund ownership for next period:

$$s_{i,t+1}^M = \frac{Q_{i,t}\phi_{i,t+1}}{\tilde{P}_{i,t}}. \tag{35}$$

A standard Taylor approximation yields $r_{i,t+1}^M \approx r_f + \phi_{i,t+1}(r_{i,t+1} - r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma_{i,t}^2$, where $\sigma_{i,t}^2 = \mathrm{Var}_t(r_{i,t+1})$ is the conditional variance of returns. Given normally distributed returns and dropping the constant terms, the fund manager's discounted payoff can be rewritten as

$$u^M = \phi_{i,t+1}(\mu_{i,t} - r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma_{i,t}^2 + \frac{1}{2}(1-\gamma_m)\phi_{i,t+1}^2\sigma_{i,t}^2, \tag{36}$$

[4]With a CRRA preference, the manager's disconted utility is $\beta\tilde{Q}_{i,t+1}^{1-\gamma_m}/(1-\gamma_m)$. Taking the first order derivative with respect to $\tilde{Q}_{i,t+1}$ gives us the marginal utility $\beta\tilde{Q}_{i,t+1}^{-\gamma_m}$. The log marginal utility is therefore $\log\beta - \gamma_m\tilde{q}_{i,t+1}$, consistent with (30).

where $\mu_{i,t} = \mathbb{E}_t[r_{i,t+1}]$ is the conditional expected return.

Fund managers face two portfolio adjustment constraints. First, managers avoid deviating too far away from targeted positions $\bar{\phi}$ due to investment mandates, risk management requirements, and index benchmarking. This constraint is captured by the portfolio adjustment cost $\frac{c_\phi}{2}(\phi_{i,t+1} - \bar{\phi})^2$, where c_ϕ controls the size of the cost. Second, executing trades is costly, leading managers to spread trades over time (Kyle, 1985). Since the model features one-period living funds, I capture this smoothing incentive by an ownership adjustment cost $\frac{c_s}{2}(s^M_{i,t+1} - s^M_{i,t})^2$, where c_s controls the cost of deviating from previous period's ownership.

Finally, the maximization problem can be written as

$$\max_{\phi_{i,t+1}} \; u^M(\phi_{i,t+1}) - \frac{c_\phi}{2}(\phi_{i,t+1} - \bar{\phi})^2 - \frac{c_s}{2}(s^M_{i,t+1}(\phi_{i,t+1}) - s^M_{i,t})^2 \tag{37}$$

subject to constraints (33) through (36). Importantly, as is analyzed in detail in Section 3.4, conditional returns and volatilities that affect payoff u^M are all functions of the choice variable $\phi_{i,t+1}$.

3.3 Residual Investors

Each firm is paired with a continuum of one-period living residual investor representing all other shareholders. Following Section 2, they solve

$$\max_{s^R_{i,t+1}} \; s^R_{i,t+1}(P_{i,t} - \tilde{P}_{i,t}) - \frac{c_r}{2}\left(s^R_{i,t+1} - s^R_{i,t}\right)^2 \tilde{P}_{i,t}, \tag{38}$$

where constant c_r controls the trading cost magnitude.

The first-order condition gives optimal share holding

$$s^R_{i,t+1} = s^R_{i,t} + \frac{P_{i,t} - \tilde{P}_{i,t}}{c_r \tilde{P}_{i,t}} \tag{39}$$

Rearranging (39),

$$\frac{s^R_{i,t+1} - s^R_{i,t}}{s^R_{i,t}} = \frac{1}{c_r s^R_{i,t}} \frac{P_{i,t} - \tilde{P}_{i,t}}{\tilde{P}_{i,t}}. \tag{40}$$

In other words, $1/(c_r s^R_{i,t})$ is the elasticity with respect to price pressure for residual investors.

3.4 Market Clearing and Price Pressure

By substituting $s^R_{i,t+1}$ and $s^M_{i,t+1}$ into the equity market clearing condition,

$$s^R_{i,t+1} + s^M_{i,t+1} = 1, \tag{41}$$

we get

$$\tilde{P}_{i,t} = \frac{1}{1 + c_r \left(s^M_{i,t} - s^M_{i,t+1} \right)} P_{i,t} \equiv \Lambda_{i,t} P_{i,t}, \tag{42}$$

where $\Lambda_{i,t}$, fluctuating around 1, denotes the degree of price pressure.

3.5 Equilibrium

Let S denote the vector of state variables $(A_t, X_{i,t}, Q_{i,t}, L_{i,t}, K_{i,t}, s^R_{i,t})$. With some abuse of notation, the recursive equilibrium is defined as

1. a cum-dividend fundamental value function for firms $V(S)$ (and hence ex-dividend value $P(S)$);
2. a set of policy functions for firms $K'(S), L'(S)$;
3. portfolio holding decisions for mutual funds $\phi'(K', L', S)$;
4. price pressure function for firms $\Lambda(S)$;

such that in each period

1. Taking $\phi'(K', L', S)$ as given, the firm chooses $K'(S)$ and $L'(S)$ to maximize $V(S)$
2. Observing K', L', taking $\Lambda(S)$ and $V(S)$ as given, mutual funds calculate expected returns and variances and choose portfolio weights $\phi'(K', L', S)$.
3. Given K', L' and $s^{M'}$, residual investors choose $s^{R'}$.
4. Markets clear for each state. Realized price pressures are consistent with $\Lambda(S)$,

$$\frac{1}{1 + c_r \left[(1 - s^R) - \frac{Q\phi'\left(K'(S), L'(S), S\right)}{\Lambda(S) V(S)} \right]} = \Lambda(S) \tag{43}$$

The timing of events within each period is as follows:

1. All shocks are realized and assets are reshuffled among newly established mutual funds.
2. Firms make investment and financing decisions.
3. Mutual funds choose portfolio weights.
4. Residual investors determine their positions.
5. Transactions are executed, markets clear, and prices are realized.

Details of the computation algorithm are provided in Appendix C. Several key features help make this system numerically tractable. First, the sequential nature means early movers only need to form beliefs about late movers' decisions rather than solving the optimization for all potential strategy combinations. Second, firms' investment and borrowing decisions determine both next-period state variables and current-period issuance and buyback decisions. This approach prevents the state space for investors, whose decisions depend on the firm's actions, from becoming unmanageable. Third, given firm and mutual fund actions, residual investors' optimal decisions have closed-form solutions that, combined with market clearing, directly yield the price pressure rule.

Mutual funds calculate the return of firm i by

$$R_{i,t+1} = \frac{\tilde{P}_{i,t+1} + O_{i,t+1}}{\tilde{P}_{i,t}}. \tag{44}$$

While at the optimum, it is still true that

$$V_{i,t} = O_{i,t} + \mathbb{E}_t[M_{t,t+1}V_{i,t+1}] \implies 1 = \mathbb{E}\left[M_{t,t+1}\frac{V_{i,t+1}}{V_{i,t} - O_{i,t}}\right] \tag{45}$$

$\mathbb{E}_t[M_{t,t+1}R_{i,t+1}] \neq 1$ due to the price pressure terms.

The equilibrium return differs from traditional dynamic firm models (e.g., Zhang, 2005), due to stock market segmentation, where each mutual fund and residual investor can only invest in their assigned firm. This segmentation is similar to the preferred-habitat literature on the term structure of interest rates (e.g., Vayanos and Vila, 2021). While residual investors follow the market-wide SDF (30), their trading is constrained by trading costs. Consequently, firm pricing reflects the baseline market-wide marginal utility adjusted by a factor measuring residual investor's marginal trading costs.

4 Calibration and Estimation

This section presents the calibration and estimation of model parameters by targeting data moments. I also employ instrumental variable estimation within the model-simulated economy to illustrate the source and magnitude of the bias from ignoring endogenous firm actions.

To maintain consistency with the empirical analysis, the model is solved at a quarterly frequency. I calculate the model-implied moments by simulating 10 samples and reporting the cross-sample average. Each sample contains 3,000 firms simulated over 1,000 quarters. To mitigate the effects of initial conditions, I discard the first 400 quarters and treat the remaining simulated data as drawn from the economy's stationary distribution.

4.1 Calibration

Table 5 reports the calibrated parameters in the baseline model. To avoid complicating the estimation with an excess of parameters, I calibrate the parameters that do not directly affect the demand and supply dynamics by first estimating them outside of the dynamic model. If a direct estimation is not possible, the parameters are set by matching some selected moments or using values reported in previous studies.

Table 5 Here

Aggregate parameters. The long-run mean of mutual fund size $\bar{q}$ is calibrated such that the average mutual fund AUM is one-third of average firm market value. The inverse of risk-free rate β is set to be 0.995 at quarterly frequency. The risk aversion parameters γ_0 and γ_1 are set to match an average aggregate stock market return of 3% per quarter and an annual Sharpe ratio of 0.35.

Mutual Funds. The persistence and volatility of idiosyncratic flow shocks are directly estimated from data. On average, mutual funds own 20% of a firm. The firm's market value is three times the size of the fund. This implies that targeted position $\bar{\phi} = \bar{s} \times (\bar{\bar{P}}/\bar{Q}) = 0.2 \times 3 = 0.6$. The risk aversion coefficient γ^M is set to 2.0, close to the median value of 2.5 estimated by Koijen (2014), and also roughly consistent with the targeted position $\bar{\phi}$[5].

[5]Absent the position adjustment costs and price impacts, the optimal position of a mutual fund with CRRA utility is $\phi = (\mu - r_f)/(\gamma\sigma^2)$, where $\mu - r_f$ is the risk premium and σ^2 is the variance of returns. With an annual risk premium of 6% and a volatility of 20%, the optimal position is 0.75 when $\gamma = 2$.

Firms. Following Zhang (2005) and Belo et al. (2019), the persistence ρ_x and the volatility σ_x of productivity shocks are set to 0.97^3 and 0.19, respectively. The curvature of the production function α, quarterly depreciation rate δ, corporate tax rate τ are standard parameters, set at 0.65, 0.03, and 0.3, respectively. Following Warusawitharana and Whited (2016), dividend tax rate τ_D is 0.15. The collateral constraint $\phi = 0.8$ and the quarterly saving-borrowing wedge $\kappa = 0.005/4$ follow Livdan et al. (2009); Belo et al. (2019); Choi et al. (2021).

4.2 Estimation

I estimate the seven key parameters that directly affect supply and demand dynamics using indirect inference.

On the firm side, the issuance cost parameters c_{h0} and c_{h1} and the repurchase cost parameter c_{b0} and c_{b1} directly determine both the average level of net equity issuance and the reaction of equity issuance to cost changes caused by price pressure. The investment adjustment cost parameter c_k governs the intertemporal Euler equation and implicitly determines both issuance and repurchase decisions.

For mutual funds, the portfolio deviation cost c_ϕ captures benchmarking intensity. In the limit where c_ϕ approaches infinity, mutual funds are fully passive so their portfolio ϕ stays constant, and all changes in s^M are driven solely by the fluctuation in their size.

Finally, for given changes in mutual fund holdings, the residual investor trading cost c_r directly controls the size of price pressure in (42). As derived in Section 3.3, the inverse of c_r represents the semielasticity with respect to price pressure.

I use eleven target moments for the indirect inference estimation.

The first five moments capture the dynamic interaction between mutual funds, residual investors, and firms. First, I target the IRF of returns at $h = 0$ and $h = 1$[6]. The initial response ($h = 0$) is directly related to the cost parameters. The return response at $h = 1$ captures dynamic adjustments. Stronger return reversals occur when initial reactions are primarily driven by price pressure or when mutual funds pay less effort to smooth their position changes. I focus on the first two periods because the model lacks mechanisms for significant long-term

[6] In the model the IRF of returns is defined as the change in log market value $\Delta \log \tilde{P}$ for easier comparison and decomposition in Section 4.4. Empirically the IRFs of $\Delta \log \tilde{P}$ and R are similar. This is because $\Delta \log \tilde{P} = \log\left(1 + \frac{\tilde{P}_{t+1} - \tilde{P}_t}{\tilde{P}_t}\right) \approx \frac{\tilde{P}_{t+1} - \tilde{P}_t}{\tilde{P}_t} = R_{t+1} - 1 - \frac{O_{t+1}}{\tilde{P}_t}$, and the payout term $O_{t+1}/\tilde{P}_t$ is an additive level not directly affected by the shock realized at t, so it drops out of the impulse response.

responses, and empirical evidence suggests that longer-term responses are not significantly different from zero. Second, I use the IRF of NEI at $h = 0$. High firm issuance and repurchase costs reduce firms' responsiveness to price pressure. Third, I target the variance of changes in mutual fund ownership share (Δs^M), which governs the extent of active portfolio rebalancing and determines the magnitude of return responses. Fourth, I target the variance of returns. Given the volatility of productivity shocks and fund flow shocks, it reveals market participants' capacity to absorb shocks.

The next six moments discipline parameters related to firm technology and financing. To match the dynamics of NEI, I calculate the mean and variance of NEI conditional on being non-negative or non-positive.[7]. I also target the regression coefficient of $SALE/K$ on I/K. A large coefficient indicates investment dynamics driven primarily by productivity shocks, while a small coefficient suggests a more important role for fund flow shocks, which affect issuance costs. Finally, I target the variance of investment rate, which directly enters financing and investment costs.

Since this model is overidentified, I construct the weighting matrix for the objective function using the influence function approach in Erickson and Whited (2002). Let x denote the data, b denote the vector of parameters, g denote the difference between data and simulated moments, and $\hat{W}$ denote the weighting matrix. The indirect inference estimator of b is then defined as the solution to the minimization of

$$\hat{b} = \arg\min_{b} g(x, b)' \hat{W} g(x, b).$$

Computational details are provided in Appendix C. Table 6 presents the estimation results.

Table 6 Here

Panel A of Table 6 reports the estimated parameters. The key parameter of interest is the trading cost for residual investors, c_r. Since residual investors on average hold 80% of a firm's shares, their elasticity with respect to price pressure, $1/(c_r s^R)$, averages approximately 1.6 across the model's stationary cross-section of s^R. This is consistent with the empirical evidence provided by Schmickler and Tremacoldi-Rossi (2022).

[7]Zero-NEI observations contribute to both averages. Since many observations are small but non-zero, this approach captures NEI levels without imposing arbitrary thresholds for "active" versus "inactive" NEI, which could introduce biases to the extensive margin of NEI. Since the firm side is overidentified, other moments help indirectly identify relevant parameters.

Panel B of Table 6 compares the moments calculated from the model with those from the data. Despite overidentification, the model fits the data well overall. The return reversal at $h = 1$ is stronger than in the data, which is expected given that one-period-living investors generate more immediate and pronounced reactions. The discrepancy is weaker in the longer horizon. For example, the cumulative return from $h = 1$ to $h = 3$ is -0.81 in the data and -1.66 in the model.

While the standard errors provide initial evidence on local identification, I further investigate the sources of identification by computing the sensitivity matrix of $\hat{b}$ to $\hat{g}(b)$ using $-(G'WG)^{-1}G'W$, where G is the Jacobian, following Andrews et al. (2017). As the raw sensitivity measures are not scale invariant, I standardize the sensitivity matrix by scaling each element with the square root of the ratio of moment variance to parameter variance to facilitate interpretation. Table 7 presents the local elasticities of parameter estimates with respect to the targeted moments, calculated at estimated parameters.

Table 7 Here

The sensitivity matrix highlights why the model's parameters must be estimated jointly. The variance of returns responds to all seven parameters (magnitudes 0.124 to 0.492), including firm-side cost parameters such as c_{b0} (0.211) and c_{b1} (-0.211); the variance of investment likewise responds to the residual investor cost c_r (-0.139) alongside the firm-side cost parameters. Fund benchmarking intensity c_ϕ and residual investor cost c_r both shape price pressure but are identified through different moments. c_r's largest loadings are on the return response (Return IRF $h = 0$: 0.643; $h = 1$: -0.446; Var(R): -0.492), with a stronger return response associated with a higher estimated c_r and thus a lower demand elasticity, while c_ϕ's largest loading is on the variance of ownership changes (Var(Δs^M): 0.650), a moment c_r barely moves (-0.148). On the firm side, the capital adjustment cost c_k and the issuance/repurchase cost levels c_{h0}, c_{b0} are most strongly identified by NEI-related moments (NEI IRF $h = 0$: c_k 0.692, c_{b0} 0.716; c_{h0}'s largest loading is Var(I/K), 0.531). These NEI moments are themselves jointly determined by all four issuance and repurchase cost parameters rather than by one side alone: the mean and variance of NEI conditional on issuing respond about as strongly to the repurchase costs c_{b0}, c_{b1} (-0.199, 0.200 and 0.469, -0.479) as to the issuance costs c_{h0}, c_{h1} (-0.187, 0.232 and 0.492, -0.705), and it is this joint response that pins down the curvature parameters c_{h1}, c_{b1} specifically through the variance of NEI conditional on issuing or repurchasing (-0.705, -0.479).

4.3 Return Response Decomposition

The observable return response to a fund flow shock combines two distinct components: the transitory price pressure wedge and the change in fundamental value it triggers. Following (42),

$$d \log \tilde{P} = d \log \Lambda + d \log P, \tag{46}$$

and Table 8 reports these two components of the initial return response. The price pressure response (1.39%) is weaker than the return response (1.67%) at $h = 0$ due to improved fundamental value following NEI responses. The gap between the observable return response and the unobservable price pressure response is the source of bias in the demand elasticity estimate. To explore heterogeneity across firms, I independently sort firms into terciles by beginning-of-period Tobin's Q and by beginning-of-period leverage, and report the four groups formed by intersecting the top and bottom terciles of each sort.

Table 8 Here

In the full sample, 17% of the initial return response is driven by improvements in fundamental value, while the remaining 83% is attributed to price pressure. There is substantial heterogeneity across firms. Consistent with the empirical findings in Table 4, firms with high Tobin's Q exhibit both stronger return responses and a larger fundamental value share.

Table 4's unconditional leverage sort looks puzzling because return and NEI responses do not move together: high-leverage firms show a stronger return but a weaker NEI response than low-leverage firms (2.243%, 0.147% vs. 1.624%, 0.229%). The model resolves this as a composition of two regimes that run in opposite directions depending on firms' growth opportunities. Among low-Q firms, high leverage is associated with a larger fundamental value response, consistent with cheap issuance being used mainly to pay down debt when investment opportunities are limited. Among high-Q firms, the pattern reverses: low leverage is associated with a larger fundamental value response, consistent with financing flexibility being used to fund expansion when investment opportunities are abundant. This composition of offsetting forces across firms with different growth opportunities is analyzed further in Section 5.

4.4 Recovering the Corrected Elasticity: An Instrumental Variable Comparison

This section delivers the paper's two headline elasticity numbers: instrumenting observed price changes, the standard approach, recovers a biased estimate of 1.3, while instrumenting price pressure directly recovers the corrected elasticity of 1.6. I estimate both demand elasticities, for residual investors and for mutual funds, using an instrumental variable approach on the model-simulated economy.

To estimate the demand elasticity for residual investors, I need a residual supply shock that is orthogonal to the residual investor demand (39). In the model, fund flow shocks are demand shocks independent of residual investor demand by construction, seemingly satisfying the traditional exclusion restriction. However, as shown in Section 4.3, this approach still yields biased results in our context because we cannot disentangle changes in firm value driven by price pressure Λ from those driven by fundamental value P, introducing inevitable omitted variable bias.

To illustrate this potential bias, I conduct instrumental variable estimation using the model-simulated economy, as reported in Panel A of Table 9. I regress changes in log ownership by residual investors against different measures of price changes, using fund flows as instruments.

Table 9 Here

In columns 1 and 2, the independent variable is changes in log market value $d\log\tilde{P}$. Without instrumenting, the estimate is positive, illustrating the traditional omitted variable bias. When firm fundamentals change, mutual funds endogenously rebalance their portfolios to achieve the optimal return-variance trade-off, so the estimate represents a weighted average of optimization responses by firms, mutual funds, and residual investors. In Column 2, using fund flow shocks as instruments yields a demand elasticity estimate of approximately 1.3.

Columns 3 and 4 use changes in log price pressure $d\log\Lambda$ as the independent variable, which is unobservable in actual data but can correctly remove the endogenous fundamental value changes. Without instruments (column 3), the estimate is already -0.6, reflecting that residual investors reduce holdings in response to positive price pressure to mitigate value deviation. With proper instrumenting, Column 4 estimates the true demand elasticity of 1.6, about 20% higher than the estimate in Column 2.

The specification in columns 3-4 remain slightly misspecified. Following (39), the correct specification should use deviations from fundamental value $(P_{i,t} - \tilde{P}_{i,t})/\tilde{P}_{i,t}$ instead of changes in price pressure. Columns 5 and 6 present the estimates using this specification. With or without instruments, the estimates are of similar magnitude and close to the estimate in Column 4, indicating that the bias from using changes in price pressure instead of the deviation measure is small.

I estimate demand elasticity for mutual funds in Panel B of Table 9. Following Koijen and Yogo (2019), the demand curve is downward sloping with respect to firm market value (rather than price pressure) because larger firms on average have lower expected returns. The demand elasticity estimate thus shows how the size premium is reflected in mutual funds' portfolio choices, which is not directly related to the price pressure. In this case, productivity shocks help firms grow but do not directly affect mutual fund demand, satisfying the exclusion restriction.

Columns 1 to 3 provide the reduced form, first stage, and IV estimates using changes in log market value. The first stage in Column 1 is biased upward. While mutual funds do cut holdings when firms grow bigger, they also increase holdings due to positive fund flow shocks, which rasies the market value through price pressure. With proper instrumenting, the IV estimate in Column 3 is -0.7, close to elasticity estimates surveyed in Gabaix and Koijen (2022). This estimate is still biased because when mutual funds adjust their holdings following productivity shocks, they also create price pressure.

Columns 4 to 6 use changes in log fundamental value, which is unobservable in actual data but can correctly remove the endogenous price pressure changes. Comparing Columns 4 and 6, the bias in the first stage is now small because biases introduced by price pressure are already removed. The elasticity estimate in Column 6 is smaller in magnitude than the estimate in Column 3, suggesting that mutual funds are more sensitive to price pressure increases than to fundamental value increases. This makes sense because the reduction in expected returns from price pressure should be sharper than that from fundamental value increases. Without additional instruments in the model, we cannot separately identify their demand elasticity with respect to price pressure as they are the ones introducing price pressure in the first place.

5 Mechanism

Section 4 established a corrected demand elasticity of 1.6, versus a biased estimate of 1.3, and showed that 17% of the initial price response reflects firms' fundamental-value improvement rather than price pressure. Section 2 showed analytically why responses to a price change should depend on its source, its size, and the firm's state; this section shows how these same three forces operate in the full dynamic model's policy functions, generating both the numbers above and the diminishing marginal price effects documented below.

5.1 Firm Optimality Conditions

The firm's investment, capital, and leverage choices are pinned down by three optimality conditions that make the price-pressure channel explicit: alongside the standard marginal costs and benefits of investing and borrowing, each condition carries a term through which firms internalize their effect on mutual fund trading and, through it, their own cost of capital. Let q and μ be the Lagrange multipliers for the capital accumulation (20) and borrowing constraint (22). The first-order conditions with respect to I, K', and L' are

$$-\frac{\partial O}{\partial I} = q \tag{47}$$

$$q = \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial K'}\right]\right\} + \mu\varphi + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial K'} \tag{48}$$

$$\mu - \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial L'}\right]\right\} = \frac{\partial O}{\partial E}\frac{\partial E}{\partial L'} + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial L'}. \tag{49}$$

The left-hand side of (47) represents the marginal cost of investment: investing more reduces current payout. The first two terms on the right-hand side are standard marginal benefits of capital accumulation. More capital increases future production capacity and relaxes the borrowing constraint. The last term reflects the novel firm-investor interaction: firms recognize that their investment decisions affect mutual fund trading through $\frac{\partial \phi'}{\partial K'}$, which in turn influences their cost of capital and payout $\frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}$.

Similarly, the left-hand side of (49) represents the marginal cost of borrowing: potentially binding borrowing constraints and future debt repayment. The marginal benefit on the right-hand side includes higher current cash flows, and the effect of borrowing on mutual fund trading and thus payout costs through $\frac{\partial \phi'}{\partial L'}$.

5.2 Equilibrium Price Formation

The full equilibrium price-formation process shows that price pressure alone does not pin down expected returns: how price pressure and expected returns co-move depends on the shape of both mutual funds' and firms' equilibrium responses, not on the magnitude of price pressure by itself. Figure 3 illustrates this equilibrium price formation process for a firm seeking to increase investment. Following backward induction[8], the top panel shows equilibrium expected returns and volatility as functions of mutual fund portfolio weights, given optimal firm decisions. When mutual funds increase weights, current price pressure rises, reducing the expected returns. For this firm, conditional volatility also decreases, creating a trade-off between lower returns and lower volatility for mutual funds. However, since the volatility variation is relatively muted, the trade-off between returns and adjustment costs in (37) likely dominates.

Figure 3 Here

Moving backward in the analysis, firm decisions directly affect conditional expected returns and volatility by changing both current and future fundamental values. These changes influence mutual funds trade-offs, which affect price pressure, and feed back into expected returns and volatility. The middle panel shows how returns and volatility depend on firm investment decisions, taking mutual fund portfolio responses as given. The dashed black line indicates current capital level, while the dotted red line shows equilibrium optimal capital for next period. Firm actions have a particularly large impact on volatility.

Considering both steps, the bottom panel plots equilibrium price pressure and mutual fund portfolio weights as functions of firm investment decisions. As shown in (42), price pressure closely tracks mutual fund holdings. However, higher price pressure does not automatically translate into lower expected returns. The range of movement in μ (the left panel in the second row) is significantly larger than the range of movement in Λ (the left panel in the bottom row), and the shape of the two curves differ significantly. This is exactly the equilibrium chain that the state- and size-dependence results below trace out across firms and shock sizes.

[8]The last step where residual investors form their portfolios follows (39).

5.3 Response to Fund Flow Shocks

The equilibrium price-formation chain traced above operates for every firm, but how strongly it operates depends on the firm's state. Figure 4 plots policy functions against idiosyncratic mutual fund size Ω, weighted by the model's stationary distribution over firm states. The Overall row shows the baseline mechanism: mutual funds cut their next-period portfolio weight ϕ' in anticipation of the price correction that a larger shock implies, but portfolio adjustment costs limit how far they can unwind, so mutual fund ownership and price pressure Λ still rise with shock size. Net equity issuance rises in tandem, since the same price pressure funds are trading against makes new equity temporarily cheap for firms to issue, and firms put the proceeds to both uses: investment I/K rises and next-period leverage L'/K' falls modestly.

Figure 4 Here

Leverage is one state variable that reshapes this response. High-leverage firms are more prone to price fluctuations and risk. This shows up jointly as a sharper cut in mutual funds' portfolio weight ϕ' (from 0.62 to 0.25, versus 0.55 to 0.41 for low-leverage firms) and a larger rise in price pressure Λ. Firms' ability to exploit that price pressure still diverges sharply by leverage, even though investment rates I/K track closely across the split. Low-leverage firms respond elastically: net equity issuance is steep and sign-switching, turning to repurchases when the fund is small and to heavy issuance once the fund is large. High-leverage firms would also like to issue as price pressure rises. But their greater exposure to price fluctuations makes issuance self-defeating: issuance costs cut further into an already-strained investment budget, and issuing itself drags Λ down more sharply for such a firm. Net equity issuance for these firms drifts only slightly above zero and stays there, even as the incentive to issue keeps rising. Leverage L'/K' correspondingly stays high, close to the model's collateral limit ($\varphi = 0.8$), rather than declining further.

The bottom row splits firms by whether they are net issuers or net repurchasers. Issuers' net equity issuance is bounded well away from zero even when the fund is at its smallest, starting near 3% of capital and rising to about 10% as fund size grows; repurchasers' net equity issuance stays flat and small throughout, never far from zero. This gap is a direct signature of the fixed issuance cost c_{h0} in the issuance cost function (25): since a firm pays this cost regardless of how much it issues, issuing only a marginal amount is never optimal, so firms either forgo issuance entirely or issue a materially sized lump. Investment tracks the same divide: issuers' I/K rises from about 7.5% to 10.5%, while repurchasers' stays flat near 2.5-3%, indicating that firms only pay the fixed cost to issue when they have real growth opportunities to fund. Leverage moves comparatively little for either group, drifting down only modestly for issuers (from about 0.28 to 0.24) and staying flat near 0.42 for repurchasers, suggesting that paring down leverage is a secondary motive relative to funding growth opportunities.

5.4 Ownership as an Endogenous State Variable

Traditional corporate finance models summarize a firm's state using variables that appear on its financial statements: capital, leverage, cash. Here, a firm's investor base, the share of its equity held by mutual funds s^M, is itself a state variable that shapes its financing environment, even though it appears on no balance sheet. Figure 5 traces out this effect, plotting policy functions against s^M, weighted by the model's stationary distribution over firm states.

Figure 5 Here

The Overall row shows the underlying mechanism. Mutual funds aim for a target portfolio weight, and when current ownership s^M is far below that target, closing even part of the gap requires a large purchase; this is what generates high price pressure Λ, which falls from about 1.05 when s^M is at its lowest to about 0.68 when s^M is at its highest, as the required adjustment shrinks. The remaining three policy functions follow directly from Λ, exactly as in the response to fund flow shocks above: with price pressure highest when s^M is lowest, equity is at its cheapest then, so net equity issuance and investment are at their highest and leverage is at its lowest; as s^M rises and price pressure fades, issuance and investment recede and firms lean relatively more on debt, so leverage rises.

Leverage again produces the sharpest cross-sectional differences, echoing the pattern above. Low-leverage firms' net equity issuance responds elastically to s^M, ranging from about 0.00 to -0.17, while high-leverage firms' stays flat and small, from about 0.01 to -0.03, the same self-defeating-issuance asymmetry documented above. Leverage follows the same logic. High-leverage firms' issuance barely moves with s^M, so they have little new equity to work with, and their leverage L'/K' sits flat near the collateral limit ($\varphi = 0.8$) throughout. Low-leverage firms issue less and repurchase more as s^M rises and price pressure fades, so they lean increasingly on debt, and their leverage climbs from about 0.10 to 0.22.

5.5 Size-Dependence: Diminishing Marginal Effects

Source and state are not the only forces shaping a shock's effects; its size matters too, because the same equilibrium forces documented above limit how far a larger shock's effects can grow. When firms face positive price pressure, they issue equity to improve fundamental values, narrowing the gap between market value and fundamental value. Simultaneously, mutual funds endogenously reduce their holdings in anticipation of the resulting price correction, further dampening price pressure directly. Consequently, even with a fixed elasticity for residual investors with respect to price pressure, larger mutual fund flow shocks do not translate proportionally into larger price fluctuations due to strategic responses from both mutual funds and firms.

These effects diminish sharply at the margin as shock size grows, and Figure 6 illustrates why, plotting the initial impulse responses of market value, price pressure, and net equity issuance under different mutual fund flow shock sizes. Panel A shows the IRFs to a one-standard-deviation shock for varying shock sizes. As shock size increases, market value responses (blue solid line) rise from 1.06% at $\sigma_{\omega i} = 0.09$ to 1.71% at the baseline, but with sharply diminishing marginal effects: moving from $\sigma_{\omega i} = 0.09$ to 0.15, from 0.15 to 0.21, and from 0.21 to 0.27 yields incremental market value increases of 0.46, 0.19, and 0.09 percentage points, respectively. Price pressure (red dashed line) rises more gradually across the same range and essentially plateaus beyond $\sigma_{\omega i} = 0.24$ (1.224% at $\sigma_{\omega i} = 0.24$ and 1.220% at $\sigma_{\omega i} = 0.27$), reflecting that strategic responses from firms and mutual funds increasingly absorb larger shocks. Panel B shows the corresponding per-unit responses normalized to the baseline shock size ($\sigma_{\omega i} = 0.21$). All three per-unit responses decline rapidly as shock size increases, confirming the self-limiting nature of the equilibrium.

Figure 6 Here

As a non-targeted check on this mechanism, Table 10 reports per-unit stock return and equity issuance responses estimated on the subsample of small empirical flow shocks: normalized coefficients are larger than one for both returns and equity issuance, consistent with the same diminishing marginal effects found in the model, though returns decay somewhat faster than equity issuance empirically.

Table 10 Here

Together, these results show that a demand shock's price impact depends on its source, its size, and the firm's state, and that firms' and investors' strategic responses are self-limiting along each dimension.

6 Counterfactuals

This section presents counterfactual analyses to evaluate how demand shocks affect firm fundamentals and capital allocation. I examine four scenarios: (1) higher mutual fund flow shock volatility ($\sigma_{\omega i}$ from 0.21 to 0.24), (2) a smaller captive mutual fund sector ($\exp(\bar{q})$ from 0.29 to 0.22), (3) a less elastic residual-investor sector (c_r from 0.878 to 1.00), and (4) lower capital adjustment costs (c_k from 0.492 to 0.45).

6.1 MPK and Capital Misallocation

Following Hsieh and Klenow (2009), Haltiwanger et al. (2018), and Baqaee and Farhi (2017) I measure capital misallocation using aggregate TFP, calculated as $\text{TFP}_{agg} = Y_{agg}/K^{\alpha}_{agg}$, where Y_{agg} and K_{agg} denote aggregate output and capital, respectively. Higher aggregate measured TFP indicates more efficient capital allocation across firms. In this exercise, I specify the firm-level production function as $Y_{i,t} = A_t^{b_i} X_{i,t} K^{\alpha}_{i,t}$, following David et al. (2022) and Choi et al. (2021), where b_i captures firm exposure to aggregate productivity shocks; the baseline sets $b_i = 1$ for all firms, and Appendix C verifies that the results are robust to letting $b_i \sim \mathcal{N}(1, 0.2)$ vary across firms. Table 11 presents the percentage changes of aggregate TFP in counterfactual specifications from the baseline.

Table 11 Here

Reducing the capital adjustment cost from 0.492 to 0.45 yields the largest effect, raising aggregate TFP by 0.14%. The three financial-side counterfactuals are an order of magnitude smaller and mixed in sign: increasing flow shock volatility by 14% (from 0.21 to 0.24) lowers aggregate TFP by 0.014%; shrinking the captive mutual fund sector (from 0.29 to 0.22) raises aggregate TFP by 0.006%; and making residual investors less elastic (c_r from 0.878 to 1.00) lowers aggregate TFP by 0.028%. The central finding is that even in a market where investor demand is inelastic and firm equity issuance responses are a first-order ingredient for identification, the real misallocation cost of financial market frictions is modest—in either direction— relative to standard real investment frictions.

Since one key source of capital misallocation in this model is the NEI responses that promote over- and under-investment, one might conjecture that stronger NEI responses to flow shocks would lead to larger investment dispersion and thus higher capital misallocation. However, this conjecture overlooks the nonlinear structure of strategic equilibrium: as Section 5 shows, per-unit NEI and price responses are larger for small shocks than for large ones, so a change in shock volatility does not translate proportionally into a change in aggregate misallocation.

6.2 Discussion

In a related setting, Choi et al. (2021) provide an important benchmark for how much capital misallocation inelastic investor demand can generate, and find substantially larger misallocation gains from reducing the magnitude of non-fundamental flow volatility by half. The difference between the two estimates reflects two modeling choices this paper makes differently: how a demand shock's source is identified, and how its size maps into investment distortion.

Both papers build on the same basic logic: inelastic demand becomes a source of misallocation when it generates a large price response to non-fundamental shocks, and firms mistake that price response for a signal about true investment opportunities. Choi et al. (2021) implement this logic by treating all investor demand unexplained by a linear Fama and French (1993) five-factor system as the non-fundamental shock, and all associated price movement as demand-driven. This paper takes a complementary approach on both fronts. First, I focus on a narrower, plausibly exogenous class of shocks, mutual fund flow shocks, rather than the full residual of a factor model. This narrower starting point is not a limit on what the framework can speak to: because firm and investor responses are disciplined directly by data, the resulting equilibrium generalizes to demand shocks of other sizes and types, as the exercise in Section 5 that varies shock size illustrates. Second, I structurally decompose the resulting price response into price pressure and the fundamental-value response it triggers, which shows that part of what looks like a demand-driven price movement is in fact a genuine, permanent increase in firm value: separating the two recovers a price-pressure elasticity of 1.6, higher than the 1.3 obtained from the observed price response alone (Table 9). Following

Koijen and Yogo (2019), the logit demand system offers an economic reason for this gap: investors reduce holdings in firms with higher market value because it predicts lower future returns from lower risk, a relationship that holds in my model for variation in fundamental value, but when two firms share the same market value with different composition, one with more fundamental value and one with more price pressure, mutual funds cut holdings more aggressively against the latter, anticipating a larger correction in expected returns. Together, a narrower shock definition and this source decomposition mean that, in this setting, a given amount of investor-demand variation maps into a smaller estimated investment distortion.

The two papers also differ in how a shock's size maps into its effect. Choi et al. (2021)'s framework does not build in a channel through which firms and investors respond more forcefully as a shock grows, so a given change in assumed noise volatility translates roughly proportionally into investment dispersion. This paper's setting features exactly such a channel: as Section 5 shows, mutual funds unwind more of their position in anticipation of a sharper price correction, and firms find issuance increasingly self-defeating as price pressure rises, so per-unit price and NEI responses shrink as shock size grows rather than staying constant. This is why, in Table 11, even a 14% increase in flow shock volatility (from $\sigma_{\omega i} = 0.21$ to 0.24) moves aggregate TFP by only 0.014%. This size-dependence is not only a feature of the model: Table 10 shows that per-unit stock return and equity issuance responses estimated on the subsample of small empirical flow shocks are larger than one, a non-targeted empirical corroboration of the same diminishing marginal effects.

A further difference between the two settings is that Choi et al. (2021) allow firms to be more heterogeneous ex ante in their exposure to aggregate productivity shocks, generating more dispersed MPK before any demand-side mechanism operates. This paper's baseline does not impose that heterogeneity, yet still matches the data's dispersion in returns, investment rates, changes in mutual fund holdings, and even the standard deviation of the level of mutual fund holdings (non-targeted, 22% in model versus 17% in data). Introducing the same kind of heterogeneity directly, letting firms' aggregate-shock exposure vary as $b_i \sim \mathcal{N}(1, 0.2)$ (Table A.3 in Appendix C), leaves the misallocation estimate essentially unchanged, indicating this modeling choice is not what drives the difference between the two papers' findings.

7 Conclusion

This paper asks how firms' strategic equity issuance in response to demand-driven price pressure biases the measurement and interpretation of investor demand elasticity, and how much real capital misallocation inelastic investor demand actually causes. Using a shift-share design built on mutual fund flow shocks, I document that a flow shock raises a firm's stock price with only partial reversal and induces equity issuance, investment, and debt paydown, concentrated among firms with high mutual fund ownership and ample growth opportunities. To interpret this issuance response, I build and estimate via indirect inference a dynamic structural model in which firms, mutual funds subject to benchmarking constraints, and residual investors facing a trading cost jointly determine market value, splitting it into a fundamental-value component and a price-pressure wedge.

The results speak to both questions in turn. On the identification side, firm issuance violates the exclusion restriction standard estimation relies on: instrumenting observed market value with flow shocks recovers a coefficient of 1.3, versus the corrected price-pressure elasticity of 1.6 once fundamental value and price pressure are separated. This corrected elasticity is itself source-, size-, and state-dependent—higher for price pressure than for fundamental value, diminishing at the margin as shocks grow, and shaped by firm leverage and growth opportunities.

On the real-effects side, these same self-limiting forces cap how much inelastic demand can distort capital allocation. Financial-side counterfactuals move aggregate TFP by less than 0.03% in either direction, an order of magnitude smaller than the 0.14% gain from a comparable reduction in real capital adjustment costs, and substantially more moderate than the misallocation Choi et al. (2021) find in a related setting: inelastic investor demand is not a first-order source of capital misallocation.

This paper's analysis leaves several directions open. It focuses on mutual funds among a broader universe of investor types; incorporating more comprehensive data on investor holdings could clarify how the mechanisms here generalize to other institutional or retail investors. It also models individual firms in isolated markets for tractability; extending the framework to aggregate quantities and prices could reveal macroeconomic implications of this interaction between price pressure and firm behavior.

This paper's broader lesson is that, in equity markets, demand elasticity may not be the primitive it is in product markets, where quantity demanded as a function of price is typically treated as a stable object shared across applications. A single elasticity number often blurs a subtle distinction: a ceteris paribus derivative holding everything but price fixed, versus a total derivative agnostic to which channel moved price. Since different counterfactuals load on different channels—assessing investment distortion, for instance, requires getting the dynamics of price pressure right—an elasticity's validity should be judged by whether it isolates the channel the intended counterfactual turns on, not estimated in isolation from one. A model built around the counterfactual of interest, estimating exactly the elasticity that counterfactual requires, may therefore travel further than a general-purpose estimate. Extending this discipline to other counterfactuals in asset markets is a natural direction for future work.

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Figures

Figure 1: Firm Responses

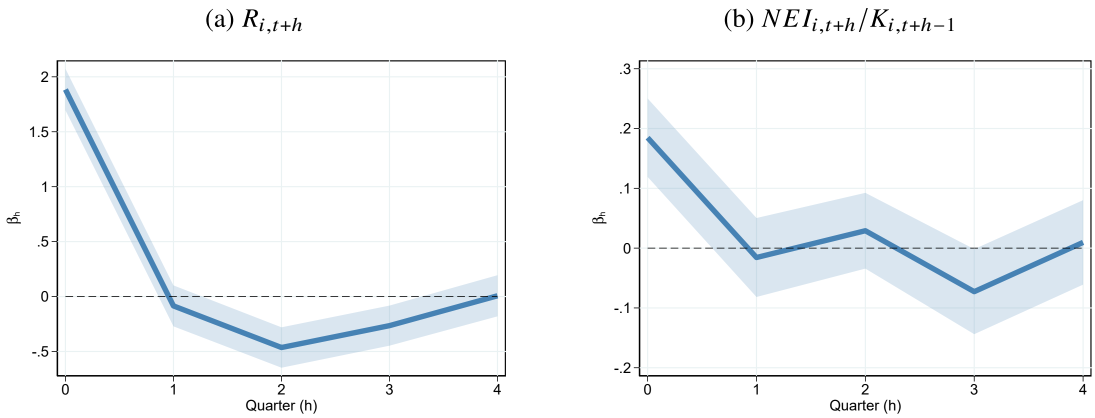
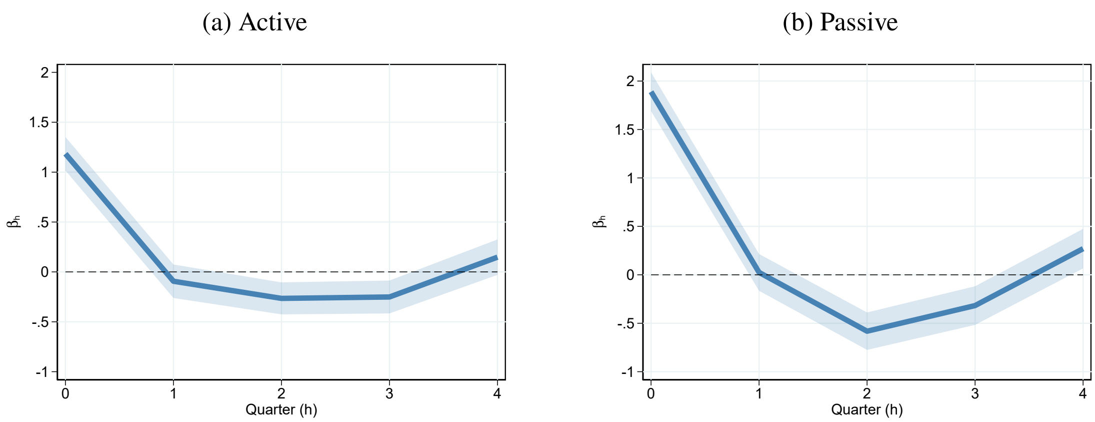


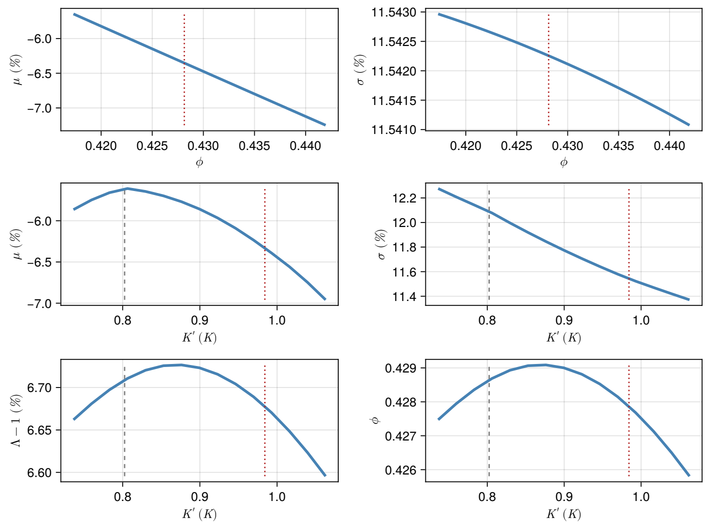
Figure 1 presents impulse response functions of returns and net equity issuance. The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005):
$\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

Figure 2: Firm Return Responses to Active and Passive Fund Flow Shocks



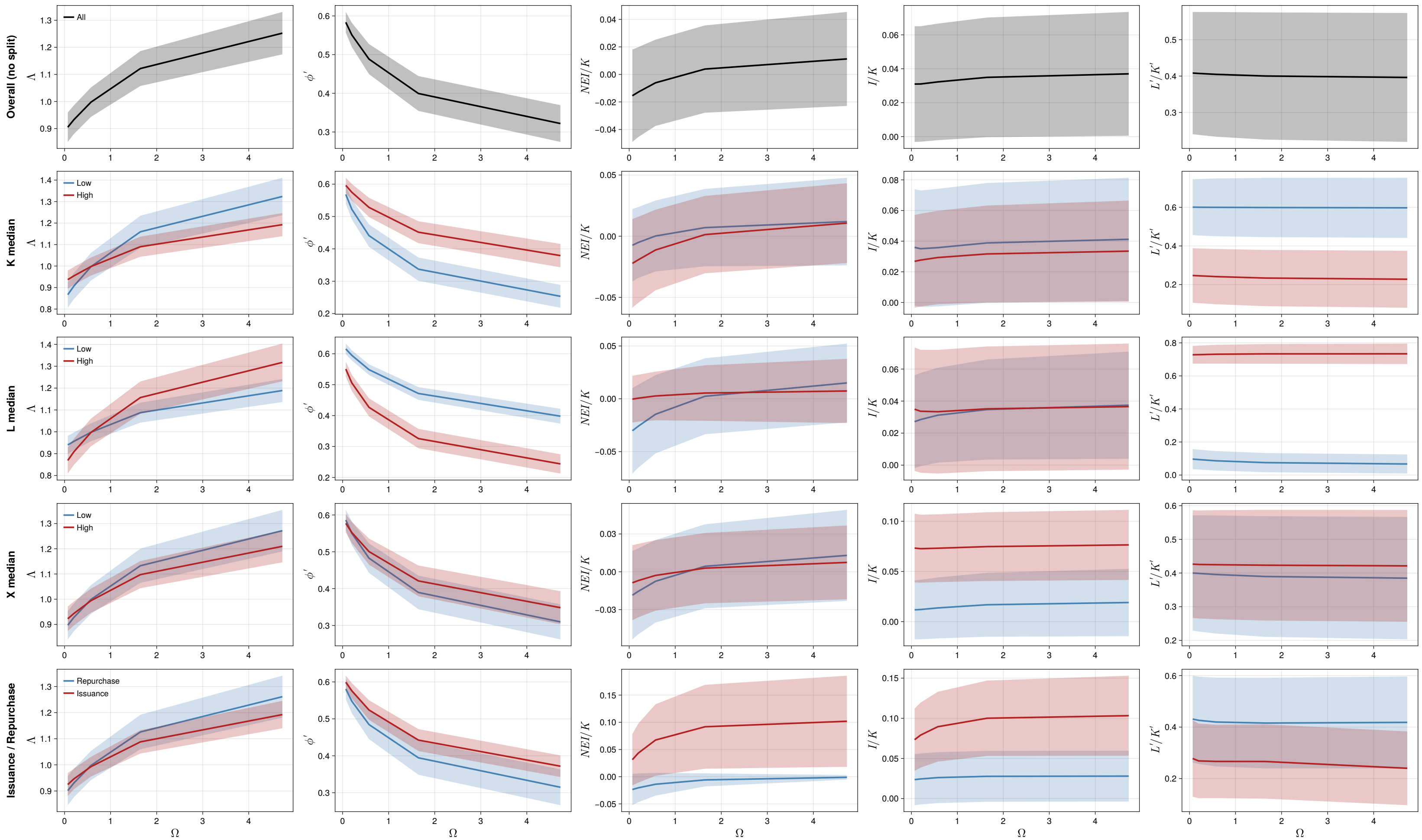
[Figure 2] presents impulse response functions of firm returns to active and passive mutual fund flow shocks ($f^T_{i,t}$ for $T \in \{\text{Active}, \text{Passive}\}$). The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005): $\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f^T_{i,t} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

Figure 3: Equilibrium Price Formation



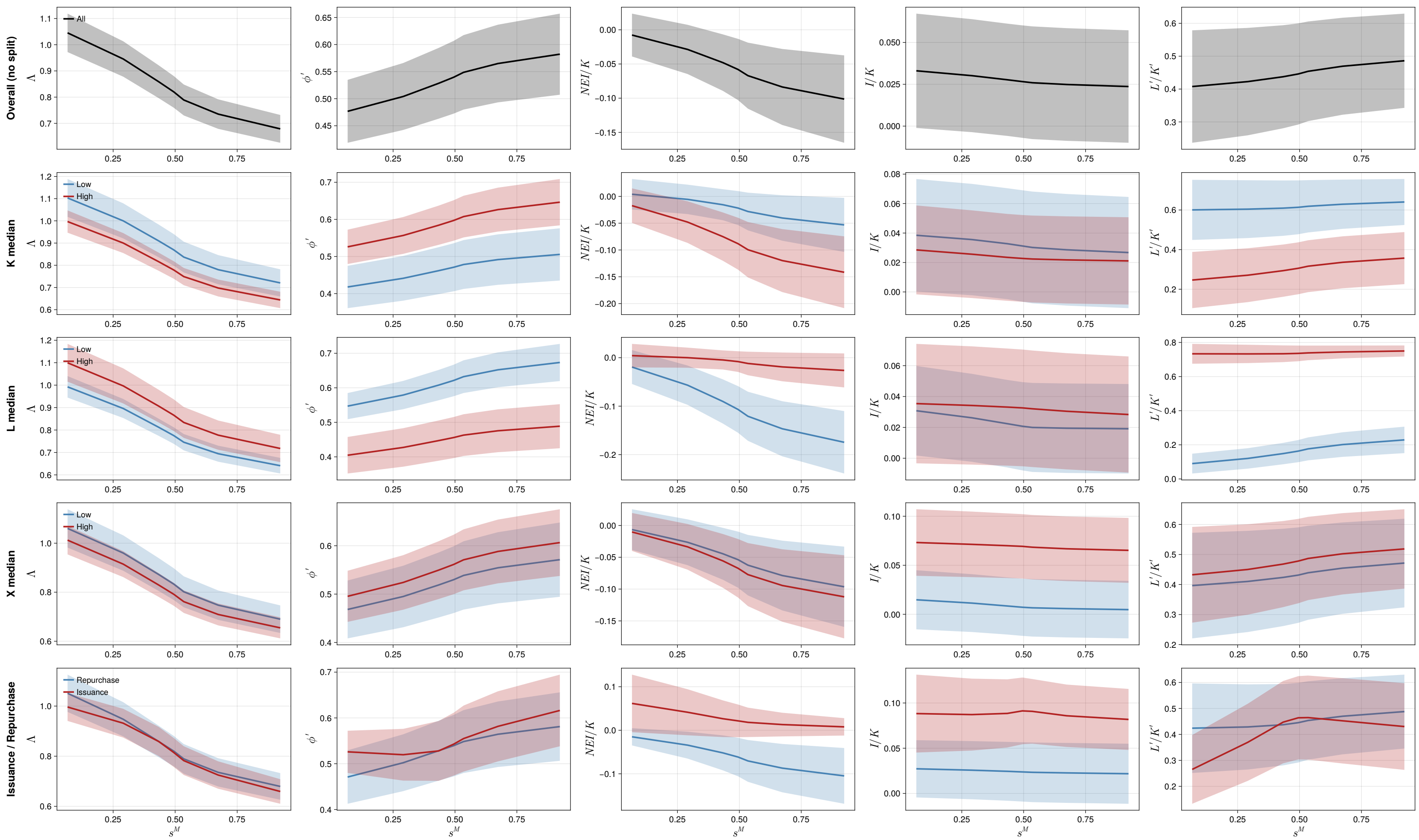
[Figure 3](#) illustrates the equilibrium price formation process for a firm seeking to increase investment. The top panel plots equilibrium expected returns and volatility as functions of mutual fund portfolio weights, given optimal firm decisions. The middle panel plots equilibrium expected return and volatility as functions of the firm's investment decisions, taking optimal mutual fund responses as given. The bottom panel plots equilibrium price pressure and mutual fund holding rules as functions of the firm investment decisions. The dotted red line indicates the mutual fund's optimal portfolio weight in the top panels, and the firm's equilibrium optimal capital for the next period in the middle and bottom panels. The dashed black line indicates the firm's current level of capital.

Figure 4: Policy Function Over Fund Size



[Figure 4](#) plots policy functions against idiosyncratic fund size Ω from the baseline model, weighted by the model's stationary distribution over states. Rows split the sample by capital K, leverage L, and firm profitability X, each at its population median (Low/High in blue/red), plus an "Overall" row (no split) and a row splitting firms by equity issuance vs. repurchase behavior (Repurchase/Issuance in blue/red). The five columns are price pressure Λ, next-period mutual fund portfolio weight ϕ', net equity issuance scaled by capital NEI/K, investment rate I/K, and next-period leverage L'/K'. Lines show the weighted mean ± 0.5 weighted standard deviation within each subsample.

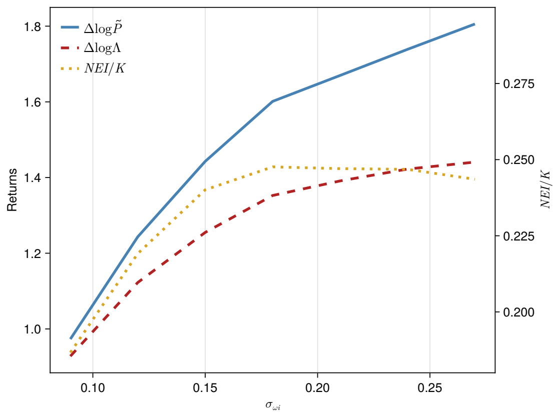
Figure 5: Policy Function Over Mutual Fund Ownership



[Figure 5](#) plots policy functions against mutual fund ownership s^M from the baseline model, weighted by the model's stationary distribution over states. Rows split the sample by capital K, leverage L, and firm profitability X, each at its population median (Low/High in blue/red), plus an "Overall" row (no split) and a row splitting firms by equity issuance vs. repurchase behavior (Repurchase/Issuance in blue/red). The five columns are price pressure Λ, next-period mutual fund portfolio weight ϕ', net equity issuance scaled by capital NEI/K, investment rate I/K, and next-period leverage L'/K'. Lines show the weighted mean ± 0.5 weighted standard deviation within each subsample.

Figure 6: Responses Under Different Shock Sizes

(a) IRF to One-Standard-Deviation Shocks



(b) Per-Unit IRF Normalized to Baseline

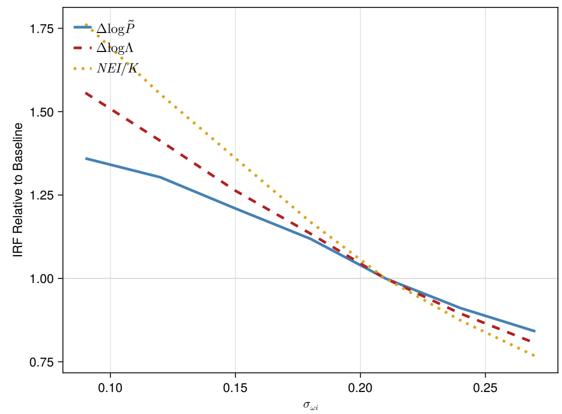


Figure 6 illustrates the initial impulse responses of market value, price pressure, and NEI under different mutual fund flow shock sizes. Panel A presents the IRFs to one-standard-deviation shocks of different sizes. Panel B presents the per-unit IRFs normalized to those under the baseline shock size. For each $\sigma_{\omega i}$, the plotted value is $\frac{\mathrm{IRF}(\sigma_{\omega i})}{\mathrm{IRF}(0.21)} \frac{0.21}{\sigma_{\omega i}}$. The blue solid line shows the market value response, the red dashed line shows the price pressure response, and the yellow dotted line shows the net equity issuance response (on the right axis in Panel A). The responses are averages across 10 simulated panels, where each panel contains 3,000 firms over 1,000 quarters, with the first 400 quarters discarded.

Tables

Table 1: Summary Statistics

	count	mean	sd	min	max
$f_{i,t}$	172,584	-0.00	1.00	-2.97	3.04
$R_{i,t}$ (%)	172,584	2.51	25.28	-66.42	89.99
$NEI_{i,t}/K_{i,t-1}$ (%)	172,584	0.94	8.45	-9.35	62.89
$s^M_{i,t}$ (%)	172,568	24.57	16.71	0.07	60.32
$dK_{i,t}$ (%, AT)	172,584	1.57	11.22	-28.45	58.70
$dK_{i,t}$ (%, PPENT)	169,147	2.20	12.61	-29.64	72.31
$dBE_{i,t}$ (%)	171,054	1.92	13.58	-57.85	81.17

Table 1 presents the summary statistics of the merged panel. $f_{i,t}$ is holding-weighted and standardized idiosyncratic mutual fund flow shocks. $R_{i,t}$ is quarterly stock return from CRSP. $NEI_{i,t}$ is Sale of Common and Preferred Stock net of Repurchase of Common and Preferred Stock and Dividends. In robustness checks, I alternatively define this variable without subtracting dividends or apply a 2% market equity filter following McKeon (2015). In the baseline, to be consistent with the model, $K_{i,t-1}$ is lagged total assets. $s^M_{i,t}$ is mutual fund holding defined as total number of shares held by mutual funds divided by total number of shares outstanding. Investment $dK_{i,t}$ is the DHS growth rate (Davis et al., 1996) of total asset. Book equity is Total Assets net of Retained Earnings and Total Liabilities. $dBE_{i,t}$ is the DHS growth rate of book equity. The sample is from 2007Q1 to 2021Q4.

Table 2: Other Firm Fundamental Responses

	$dBE_{i,t}$	$GEI_{i,t}/K_{i,t-1}$	$dCash_{i,t}$	$dK_{i,t}$ (PPENT)	$dK_{i,t}$ (AT)	ΔLeverage$_{i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
$f_{i,t}$	0.132***	0.157***	0.450***	0.209***	0.238***	-0.034***
	(3.15)	(4.70)	(2.88)	(4.26)	(5.27)	(-2.60)
Observations	157092	158593	158270	155387	158605	156590
R-Squared	0.064	0.263	0.120	0.208	0.115	0.078
Time	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Clustered By	Firm	Firm	Firm	Firm	Firm	Firm

Table 2 presents regression coefficients from $\Delta y_{i,t} = \alpha_i + \delta_t + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{m,t}$. Dependent variables include the DHS growth rate of book equity, cash, physical capital (PPENT), total assets (AT), and leverage. Gross equity issuance is defined as Sale of Common and Preferred Stock when it is larger than 2% of lagged market equity following McKeon (2015), and zero otherwise. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). All dependent variables are in percentage points. Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 3: Initial Return and Issuance Response, Positive and Negative Shocks

	$R_{i,t}$ (%)		$NEI_{i,t}/K_{i,t-1}$ (%)	
	(1)	(2)	(3)	(4)
$f_{i,t} \times 1\{f \geq 0\}$	2.111***	1.949***	0.219***	0.221***
	(13.03)	(11.82)	(3.67)	(3.68)
$f_{i,t} \times 1\{f < 0\}$	1.442***	1.821***	0.108**	0.149***
	(10.53)	(12.54)	(2.13)	(2.91)
Observations	216248	158605	216248	158605
R-Squared	0.249	0.307	0.249	0.298
Time	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Clustered By	Firm	Firm	Firm	Firm

Table 3 presents regression coefficients from $\Delta y_{i,t} = \alpha_i + \delta_t + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{m,t}$. $\mathbb{1}\{f \geq 0\}$ ($\mathbb{1}\{f \geq 0\}$) is a dummy variable that equals one when the flow shock is greater than or eqaul to (smaller than) 0. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 4: Initial Return and Issuance Response, Subsamples

	(1) $R_{i,t}$ (%)	(2) $NEI_{i,t}/K_{i,t-1}$ (%)	(3) $R_{i,t}$ (%)	(4) $NEI_{i,t}/K_{i,t-1}$ (%)
Panel A: By Mutual Fund Ownership				
	High		Low	
$f_{i,t}$	3.392***	0.195***	1.303***	0.145***
	(20.75)	(4.03)	(10.73)	(3.35)
Observations	79567	79567	72177	72177
R-Squared	0.376	0.264	0.289	0.339
Panel B: By Q				
	High		Low	
$f_{i,t}$	2.181***	0.318***	1.440***	0.006
	(14.77)	(4.59)	(10.79)	(0.30)
Observations	71570	71570	73015	73015
R-Squared	0.320	0.368	0.373	0.297
Panel C: By Leverage				
	High		Low	
$f_{i,t}$	2.243***	0.147***	1.624***	0.229***
	(14.70)	(3.66)	(12.33)	(4.41)
Observations	72433	72433	76946	76946
R-Squared	0.361	0.344	0.301	0.329
Time	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Clustered By	Firm	Firm	Firm	Firm

Table 4 presents regression coefficients from $\Delta y_{i,t} = \alpha_i + \delta_t + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{m,t}$. I sort firms by lagged mutual fund ownership, Tobin's Q, or leverage into two equal-sized groups within each Fama–French 12 industry. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 5: Calibrated Parameters

Panel A: Aggregate		
Long-run mean of fund size	$\exp(\bar{q})$	0.29
Persistence of aggregate productivity shocks	ρ_a	0.95
Volatility of aggregate productivity shocks	σ_a	0.006
Inverse of risk-free rate	β	0.995
Risk aversion, constant component	γ_0	30
Risk aversion, time-varying component	γ_1	-100
Panel B: Mutual Funds		
Persistence of idiosyncratic flow shocks	ρ_ω	0.98
Volatility of idiosyncratic flow shocks	σ_ω	0.21
Risk aversion	γ^M	2.0
Targeted portfolio position	$\bar{\phi}$	0.60
Panel C: Firm		
Persistence of idiosyncratic productivity shock	ρ_x	0.97^3
Volatility of idiosyncratic productivity shock	σ_x	0.19
Long-run mean of idiosyncratic productivity shock	$\bar{x}$	-3.08
Decreasing returns to scale	α	0.65
Depreciation rate	δ	0.03
Corporate tax rate	τ	0.3
Dividend tax rate	τ_D	0.15
Collateral constraint	φ	0.8
Saving-borrowing rate wedge	κ	0.005/4

Table 5 presents the calibrated parameters for the baseline model.

Table 6: Identified Parameters

Panel A: Parameter Estimates		
Parameter	Symbol	Value
Fixed issuance costs	c_{h0}	0.032
		(0.008)
Fixed buyback costs	c_{b0}	0.010
		(0.001)
Variable issuance costs	c_{h1}	0.552
		(0.072)
Variable buyback costs	c_{b1}	1.139
		(0.074)
Fund target deviation costs	c_{ϕ}	2.391
		(0.085)
Residual investor trading costs	c_r	0.878
		(0.031)
Capital adjustment costs	c_k	0.492
		(0.001)

Panel B: Targeted Moments		
Moment	Data	Model
Return IRF, $h = 0$ (%)	1.81	1.67
Return IRF, $h = 1$ (%)	0.11	-1.32
NEI IRF, $h = 0$ (%)	0.28	0.25
SD(Δs^M) (%)	3.40	3.26
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \geq 0]$ (%)	3.57	1.62
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \leq 0]$ (%)	-0.83	-1.02
SD($NEI_t/K_{t-1} \mid NEI_t \geq 0$) (%)	14.56	6.36
SD($NEI_t/K_{t-1} \mid NEI_t \leq 0$) (%)	1.65	1.68
β(SALE/K, I/K)	0.15	0.15
SD(R) (%)	23.87	16.84
SD(I/K) (%)	7.38	6.56

Table 6 presents estimated parameters and compares the moments from the simulated economy with data. Standard errors are in parentheses. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded. Data moments are from 2007Q1 to 2021Q4.

Table 7: Sensitivity of Parameters With Respect to Moments

	c_{h0}	c_{b0}	c_{h1}	c_{b1}	c_{ϕ}	c_r	c_k
Return IRF, $h = 0$ (%)	-0.018	-0.088	0.052	0.111	0.486	0.643	0.138
Return IRF, $h = 1$ (%)	-0.023	0.160	-0.094	-0.173	-0.351	-0.446	-0.017
NEI IRF, $h = 0$ (%)	-0.358	0.716	-0.364	-0.691	0.058	0.053	0.692
Var(Δs^M) ($\times 10^4$)	-0.109	-0.015	0.017	0.022	0.650	-0.148	0.183
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \geq 0]$ (%)	-0.187	-0.199	0.232	0.200	0.020	0.056	-0.305
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \leq 0]$ (%)	0.029	0.082	0.000	-0.088	-0.009	-0.013	-0.112
Var($NEI_t/K_{t-1} \mid NEI_t \geq 0$) ($\times 10^4$)	0.492	0.469	-0.705	-0.479	-0.042	-0.073	-0.183
Var($NEI_t/K_{t-1} \mid NEI_t \leq 0$) ($\times 10^4$)	0.044	0.187	0.011	-0.211	-0.026	-0.022	-0.117
β(SALE/K, I/K)	0.033	-0.052	0.023	0.051	0.018	0.031	-0.043
Var(R) ($\times 10^4$)	-0.453	0.211	0.124	-0.211	-0.346	-0.492	0.300
Var(I/K) ($\times 10^4$)	0.531	0.123	-0.444	-0.142	-0.037	-0.139	-0.364

Table 7 presents local elasticity of parameters (columns) with respect to moments (rows) at the estimated parameter values. The matrix is calculated following Andrews et al. (2017). Each element is scaled by the square root of the ratio of the moment variance to the parameter variance. Elasticities are calculated using central finite difference with a step size equal to 0.01 of the estimated parameter value. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded.

Table 8: Decomposition of Return Impulse Responses

	Full Sample	Low Q		High Q	
		Low Lev.	High Lev.	Low Lev.	High Lev.
Return	1.67	1.37	1.63	1.97	1.82
Price Pressure	1.39	1.19	1.37	1.53	1.50
Share (%)	*83*	*87*	*84*	*78*	*83*
Fundamental Value	0.28	0.18	0.26	0.44	0.32
Share (%)	*17*	*13*	*16*	*22*	*17*

Table 8 presents the decomposition of return impulse responses estimated from the baseline model. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded. Firms are sorted into terciles by beginning-of-period Tobin's Q and by leverage; the table reports the four groups formed by intersecting the top and bottom terciles of each sort.

Table 9: Instrumental Variable Estimation

Panel A: Residual Investor Elasticity						
	$d\log s^R$					
	(1)	(2)	(3)	(4)	(5)	(6)
$d\log \tilde{P}$	0.038	-1.305				
	(0.001)	(0.004)				
$d\log \Lambda$			-0.644	-1.567		
			(0.002)	(0.002)		
$(P-\tilde{P})/\tilde{P}$					-1.893	-1.609
					(0.004)	(0.002)
Estimator	OLS	IV	OLS	IV	OLS	IV

Panel B: Mutual Fund Elasticity						
	$d\log s^M$	$d\log \tilde{P}$	$d\log s^M$	$d\log s^M$	$d\log P$	$d\log s^M$
	(1)	(2)	(3)	(4)	(5)	(6)
$d\log \tilde{P}$	-0.174		-0.736			
	(0.004)		(0.003)			
$d\log X$		0.264			0.289	
		(0.000)			(0.000)	
$d\log P$				-0.562		-0.672
				(0.003)		(0.002)
Estimator	OLS	OLS	IV	OLS	OLS	IV

Table 9 presents instrumental variable estimations for the demand elasticity in simulated samples. Standard errors are clustered at firm level. The coefficients from the simulated sample are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded. Standard errors are in parentheses.

Table 10: Stock Return and Equity Issuance Responses to Small Fund Flow Shocks

	$R_{i,t}$ (%)		$NEI_{i,t}/K_{i,t-1}$ (%)	
	(1)	(2)	(3)	(4)
$f_{i,t}$	2.659***	3.150***	0.234***	0.205***
	(15.63)	(16.46)	(3.89)	(3.43)
Observations	127162	94005	127162	94005
R-Squared	0.232	0.299	0.284	0.334
Time	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Clustered By	Firm	Firm	Firm	Firm
Baseline coeff.	1.806	1.884	0.168	0.185
Normalized coeff.	1.472	1.672	1.389	1.112

Table 10 reports per-unit stock return and equity issuance responses to fund flow shocks, estimated on the subsample of small shocks. The sample is restricted to firm-quarters where $|f_{i,t}| \leq 1$ baseline standard deviation; within this restricted sample, the standard deviation of $f_{i,t}$ is approximately half that in the full sample. The coefficient on $f_{i,t}$ is the raw OLS estimate from the restricted subsample, in the original units (percent). The "Baseline coeff." row reports the corresponding full-sample coefficient for reference. The "Normalized coeff." row reports the ratio of the restricted-sample coefficient to the baseline coefficient, so that a value of 1.0 indicates the per-unit response in the restricted sample equals the full-sample estimate. All regressions include firm and time fixed effects with standard errors clustered by firm. Columns (2) and (4) additionally control for $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 11: Capital Misallocation: Percentage Change from Baseline

Param	Base	CF	Description	Agg TFP	SD(NEI/K)	SD(I/K)	SD(R)
$\sigma_{\omega i}$	0.210	0.240	High flow vol.	-0.014	0.421	-0.055	0.695
$\exp(\bar{q})$	0.290	0.220	Small fund sector	0.006	0.202	-0.388	0.508
c_h	0.878	1.000	High capacity constr.	-0.028	-0.115	0.330	-0.633
c_k	0.492	0.450	Low capital adj. cost	0.142	0.950	4.759	0.119

Table 11 presents the percentage changes of some moments from the baseline model. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded.

A Empirical Appendix

A.1 Variable Definition

Variables from CRSP/Compustat Merged Database

I retrieve monthly returns `ret` from CRSP, adjust for delisting returns following Bali et al. (2017), and subtract the risk-free rate. I then aggregate the excess returns to quarterly level. Number of shares outstanding is the last observation of `shrout` for each quarter. Market capitalization is calculated as the absolute value of the product of `shrout` and `altprc`.

In the baseline results, following the convention in literature, NEI is defined as `sstk` - `prstkcy` - `dvy`. For robustness, I alternatively define NEI as `sstk` - `prstkcy` to exclude effects of dividends.

I include some other firm outcome variables. dBE is the DHS growth rate of `at` - `re` - `lt`. GEI/K is `sstk` divided by lagged `at` when `sstk` is larger than 2% of the market capitalization. $dCash$ is the DHS growth rate of `che`. $dK(PPENT)$ is the DHS growth rate of `ppent`. ΔLeverage is the first difference of leverage, which is defined as `dltt` + `dlc` scaled by `at`. Tobin's Q is defined as (market capitalization + `at` - texttttceq) / `at`.

I include portfolio-level characteristics to predict mutual fund flows. Profitability is defined as (`sale` - `cogs` - `xsga` - `xint`)/Book Equity, where Book Equity is `seq` + `ceq` + `pstk` + `at` - `lt` + `txditc` - `pstk`. Log book-to-market ratio is the logarithm of Book Equity divided by market capitalization. Log market size is the logarithm of market capitalization. Log asset growth is the log difference of `at`. Market beta is calculated from rolling window regressions estimated using monthly return data. For each firm, I regress excess returns on excess market returns. The window size is 60 months and the firm must have at least 48 quarters of observations. Risk-free rate and market return is from Kenneth French's Data Library.

CRSP Survivor-Bias-Free US Mutual Fund Database

Mutual fund excess return is calculated at monthly level by subtracting market return from `mret` and then aggregated to quarterly level. Mutual fund size is `mtna`. I use `percent_tna` as weights to aggregate firm characteristics to portfolio level. I use lagged `nbr_shares` as weights to aggregate mutual fund flow shocks to firm level. In rare cases where the last observation of `nbr_shares` is missing, I use information up to two quarters to fill the missing values. Mutual fund ownership s^M is the sum of `nbr_shares` across mutual funds divided by `shrout` from CRSP.

Figure A.1 plots the coefficients estimated from the rolling window regressions. Consistent with the literature, lagged flows and excess returns have significant predictive power. However, the coefficients are time-varying. The coefficients on portfolio characteristics are also significant and time varying.

Figure A.1: Coefficients of Rolling Window Regressions

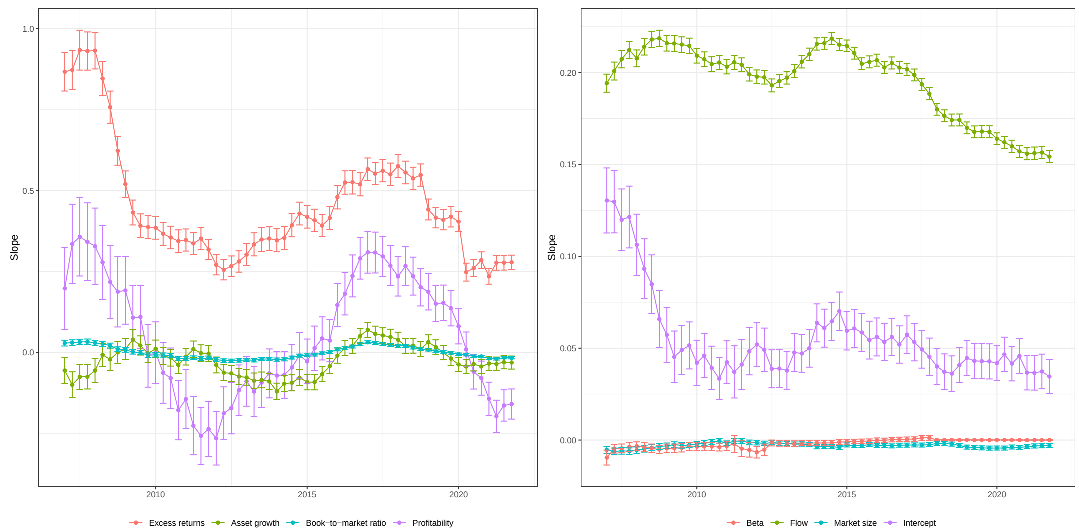


Figure A.1 presents coefficients estimated from regressing fund flows on lagged flow, fund excess returns, and lagged portfolio characteristics, including log market equity, book-to-market ratio, profitability, investment, and market beta. Each regression is performed on a panel with 16 quarters. All regressors are winsorized at 1% level.

A.1.1 Active and Passive Funds

In CRSP mutual fund database, the variable `index_fund_flag` identifies if a fund is an index fund. If a fund is a pure index fund (`index_fund_flag==D`), an index-based fund (`index_fund_flag==B`), or an index fund enhanced (`index_fund_flag==E`), I classify it as passive. Further, following Appel et al. (2016), I classify a fund as passively managed if its name includes a string that identifies it as an index fund. The strings include *"Index", "Idx", "Indx", "Ind ", "Russell", "S & P", "S and P", "S&P", "SandP", "SP", "DOW", "Dow", "DJ", "MSCI", "Bloomberg", "KBW", "NASDAQ", "NYSE", "STOXX", "FTSE", "Wilshire", "Morningstar", "100", "400", "500", "600", "900", "1000", "1500", "2000", "5000".*

A.2 Robustness

A.2.1 Endogeneity Concerns of Mutual Fund Flow Shocks

In shift-share instrument settings, validity can be established through either exogenous shares (Goldsmith-Pinkham et al., 2020) or exogenous shifts (Borusyak et al., 2018, 2025). I provide supporting evidence for both.

Share-based validity. Under the share-based design (Goldsmith-Pinkham et al., 2020; Chaudhary et al., 2022), identification requires that the lagged ownership shares $S_{i,m,t-1}$ in equation (2) are not themselves allocated in anticipation of firm investment and financing decisions. The concern is that fund managers with superior information or market-timing skills may identify promising sectors or characteristic-based trends—for instance, an anticipated industry-wide expansion such as the rise of electric vehicles, or an upcoming revival of value stocks—and overweight firms in those buckets ahead of the subsequent surge in firm investment and equity issuance. Table A.1 addresses this. Columns (1)–(2) use Fama–French 12 industry-by-quarter fixed effects, absorbing any time-varying sector-level tilt in fund portfolios. Columns (3)–(8) replace time fixed effects with size-decile-, book-to-market-decile-, and profitability-decile-by-quarter fixed effects, addressing analogous tilts based on firm characteristics. The coefficients on $f_{i,t}$ remain stable across all columns for both returns and net equity issuance.

Table A.1: Robustness: Characteristic-by-Time Fixed Effects

	Industry		Size		BM		Profitability	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$
$f_{i,t}$	1.556***	0.167***	1.753***	0.152***	1.764***	0.154***	1.806***	0.167***
	(16.20)	(4.97)	(17.51)	(4.50)	(18.18)	(4.50)	(19.26)	(5.19)
Obs.	158605	158605	158605	158605	158605	158605	158605	158605
R^2	0.342	0.304	0.321	0.335	0.321	0.304	0.333	0.325
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm

Table A.1 presents regression results from $\Delta y_{i,t} = \alpha_i + \delta_{j,t} + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{i,t}$, where $\delta_{j,t}$ denotes characteristic-by-time fixed effects. Columns (1)–(2) use Fama–French 12 industry-by-quarter fixed effects; columns (3)–(4) use size-decile-by-quarter fixed effects; columns (5)–(6) use book-to-market-decile-by-quarter fixed effects; columns (7)–(8) use profitability-decile-by-quarter fixed effects. Both dependent variables are in percentage points. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$, and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by assets, and investment). Standard errors are clustered at the firm level. The sample is from 2007Q1 to 2021Q4.

Following the reasoning in Chaudhary et al. (2022), Figure A.2 addresses the possibility of firm-specific anticipatory allocation. I reconstruct firm-level shocks using shares lagged four quarters, $f^{(4)}_{i,t} = \sum_m S_{i,m,t-4} f_{m,t} / \sum_m S_{i,m,t-4}$, ensuring that portfolio weights predate any near-term repositioning toward specific firms. The impulse responses for returns and net equity issuance are qualitatively unchanged, though the point estimates are slightly attenuated. This is expected: four-quarter-old weights are a noisier measure of contemporaneous fund exposure, as funds rebalance their portfolios over intervening quarters, introducing classical measurement error that biases the estimates toward zero.

Shift-based validity. The concern is that fund flow shocks $f_{m,t}$ reflect anticipated firm fundamentals

Figure A.2: Firm Responses: Four-Quarter Lagged Share Weights

(a) $R_{i,t+h}$ (b) $NEI_{i,t+h}/K_{i,t-1}$

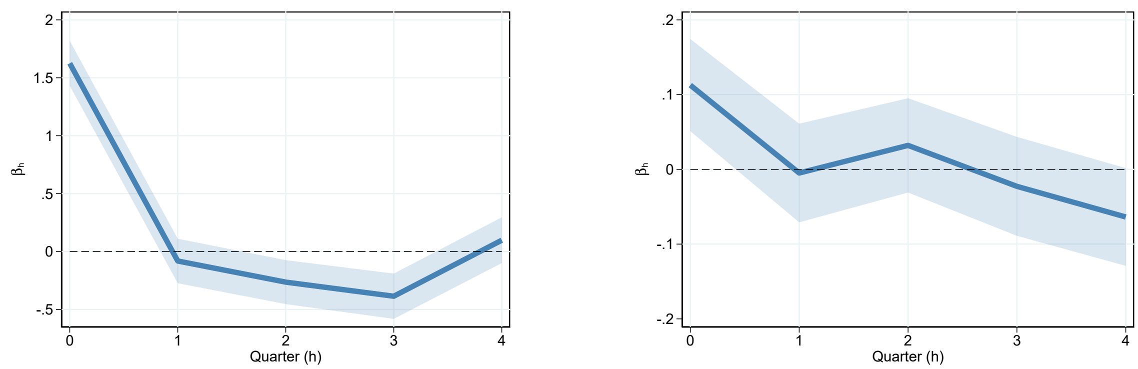


Figure A.2 presents impulse response functions of returns and net equity issuance. The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005): $\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t}^{(4)} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The firm-level shock is constructed using shares lagged four quarters as weights, $f_{i,t}^{(4)} = \sum_m S_{i,m,t-4} f_{m,t} / \sum_m S_{i,m,t-4}$, where $S_{i,m,t}$ denotes the number of shares of firm i held by fund m at time t. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}^{(4)}$, $s_{i,t-4}^{M}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

rather than exogenous demand shocks, even after controlling for observable fund and portfolio characteristics. I address three specific concerns.

First, retail investors may anticipate factor-level performance and direct flows into funds with corresponding exposures. For instance, they might flow into industry funds ahead of a sectoral expansion, or into value funds ahead of a value-factor recovery, making fund flows endogenous to future fundamentals along those dimensions. Table A.1 addresses this by replacing time fixed effects with factor-group-by-quarter fixed effects: Fama–French 12 industry, size decile, book-to-market decile, and profitability decile. Identification then comes entirely from within-group variation. The equity issuance coefficients are stable across all columns. The return coefficient attenuates mildly in the industry specification (1.556 versus 1.9% in the baseline), consistent with cross-asset substitution within industries documented by Haddad et al. (2021) and Chaudhary et al. (2022); the characteristic-based columns show minimal attenuation. Since this paper targets the total price response to fund flow shocks, not the within-group net-of-substitution effect, the baseline specification remains appropriate.

Second, fund flows may be driven by the salient performance of superstar firms within an industry rather than fund-idiosyncratic exogenous demand shocks. Table A.2 re-estimates the baseline after removing the top $N \in \{2, 10, 20\}$ largest firms by market capitalization within each Fama–French 12 industry-quarter. Both the return and equity issuance coefficients remain stable and highly significant across all columns.

Table A.2: Robustness: Excluding the Largest Firms Within Each Industry

	Top 2		Top 10		Top 20	
	(1) $R_{i,t}$	(2) $NEI_{i,t}/K_{i,t-1}$	(3) $R_{i,t}$	(4) $NEI_{i,t}/K_{i,t-1}$	(5) $R_{i,t}$	(6) $NEI_{i,t}/K_{i,t-1}$
$f_{i,t}$	1.863***	0.185***	1.805***	0.187***	1.735***	0.189***
	(19.29)	(5.50)	(18.48)	(5.49)	(17.51)	(5.44)
Observations	157471	157471	152425	152425	146547	146547
R^2	0.307	0.298	0.306	0.297	0.306	0.297
Time	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Clustered	Firm	Firm	Firm	Firm	Firm	Firm

Table A.2 replicates the baseline regression on subsamples that exclude the top N largest firms by market capitalization within each Fama–French 12 industry-quarter, for $N \in \{2, 10, 20\}$, addressing the concern that fund flows are driven by the salient performance of the largest firms within industries. Both dependent variables are in percentage points. Control variables include lagged dependent variable, lagged flow shock, mutual fund ownership, and lagged firm characteristics (log market equity, log book-to-market ratio, cash-to-assets ratio, profitability, and log asset growth). All regressions include firm and quarter fixed effects. Standard errors are clustered at the firm level. The sample is from 2007Q1 to 2021Q4.

Third, investors might observe firm actions before flow shocks and allocate capital accordingly. To address this concern, I estimate the baseline specification for sales, profitability, investment, and leverage over quarters $h = -4, \dots, -1, 0, 1, \dots, 4$ relative to the flow shocks. Sales and profitability capture top-line and bottom-line performance, investment captures forward-looking decisions, and leverage captures capital structure decisions. Figure A.3 shows no significant pre-trend in any variable, ruling out systematic anticipatory capital allocation ahead of flow shocks.

A.2.2 Alternative Definition of NEI

Figure A.4 presents the impulse response from estimating (3). The dependent variable is defined as `sstk - prstkcy`. The result is very similar to the baseline results in Figure 1. It is to be expected since the variation of dividend in the short term is small.

Figure A.3: Pre-event Firm Actions

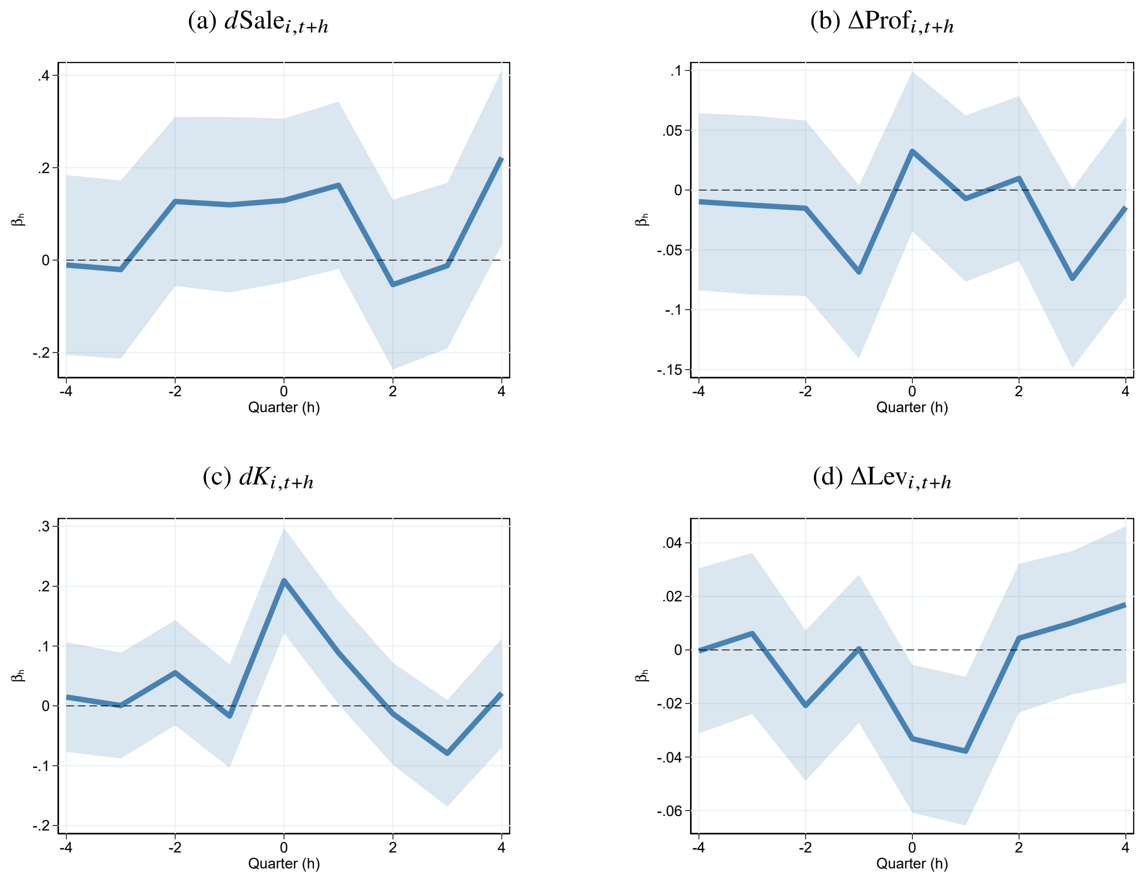


Figure A.3 presents impulse response functions of sales growth, profitability, asset growth, and leverage. The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005):
$\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-(h+1)} + \varepsilon_{m,t+h}$ for $h = -4, \dots, -1, 0, 1, \dots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-(h+1)}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ if $h \geq 0$, $\Delta y_{i,t-(h+1)}$, $f_{i,t-(h+1)}$, $s^M_{i,t-(h+1)}$ if $h < 0$. and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

A.3 Fund Flow Shocks

A.3.1 Fund Portfolio Responses

As an additional check to the results presented in Figure 1, I check whether mutual funds scale up their positions on firms they are currently holding. To that end, let $\phi_{i,m,t}$ denote the portfolio weight of mutual fund m in firm i, and $I_{m,t}$ denote the set of firms held by mutual fund m, I calculate the value-weighted increase in their current portfolio $\Delta \log S_{m,t}$ by

$$\Delta \log S_{m,t} = \sum_{i \in I_{m,t} \cap I_{m,t-1}} \phi_{i,m,t-1} (\log S_{i,m,t} - \log S_{i,m,t-1}) \tag{A.1}$$

Figure A.4: Firm NEI (Excl. Dividends) Response

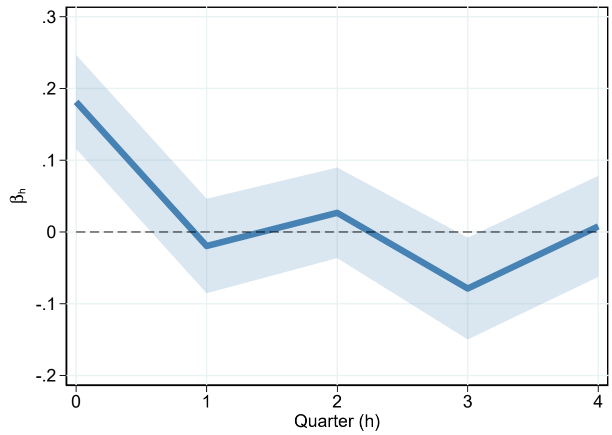


Figure A.4 presents impulse response functions of net equity issuance (excluding dividends), The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005): $\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

I estimate the following local projection regression:

$$\Delta \log S_{m,t+h} = \alpha_m + \delta_t + \beta_h f_{m,t} + X_{i,t-1} + \varepsilon_{m,t+h} \text{ for } h = 0, 1, \ldots, 4, \tag{A.2}$$

where α_m and δ_t are mutual fund and time fixed effects, and control variables $X_{i,t-1}$ include lagged portfolio weight change and flow shocks. β_h can be interpreted as the $h-$quarter ahead impulse response from the mutual fund flow shocks. The coefficients are plotted in Figure A.5. A one percent unexpected increase in fund size leads funds to increase holdings in their current portfolio by 0.56 percent on impact. Consistent with the results on Δs^M at firm level, where mutual fund ownership rises 1.4% on impact, mutual funds continue gradually raising positions in existing holdings over several quarters. Four quarters after the shock, the cumulative increase reaches 0.95 percent. In other words, while pass-through is incomplete initially, it accumulates to nearly complete over time, rather than adjusting in one step. This gradual buildup, at both the fund and firm level, suggests that mutual funds aim to avoid price pressure and prefer a gradual adjustment.[9]

This pattern has important implications. First, assuming full pass-through on impact would overestimate price impact and underestimate demand elasticity. Second, the near-complete pass-through after several quarters supports the validity of the share-based identification design, as it

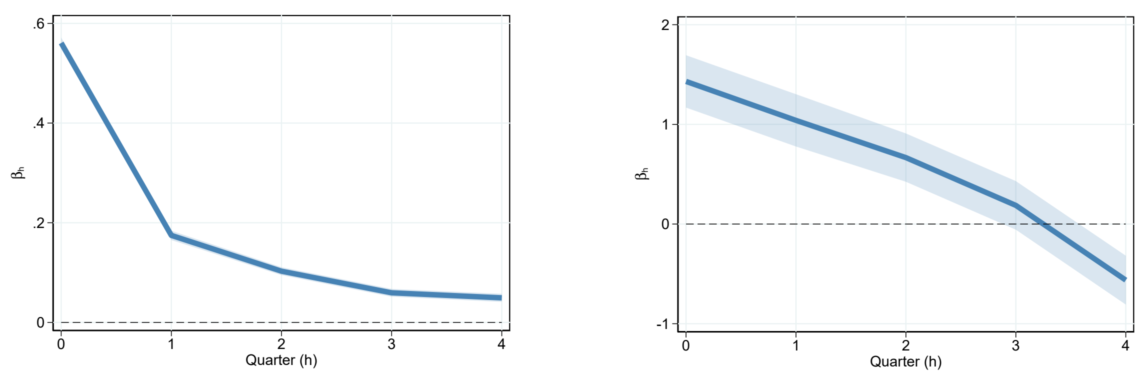
[9] Note that this is not due to persistent $f_{i,t}$ since the AR(1) coefficient of $f_{i,t}$ estimated following Han and Phillips (2010) is only 0.017. Also, $f_{i,t-1}$ is always controlled for in the regressions.

suggests funds do not systematically treat firms in their portfolios differently. Third, these dynamics underscore the importance of incorporating dynamic trading behavior into theoretical models.

Combining this pass-through evidence with the price response documented in Figure 1, I compute a simple back-of-the-envelope demand elasticity. Following Gabaix and Koijen (2022), the price impact, defined as the percentage price change divided by the percentage change in mutual fund holdings (Δs^M, 1.4% on impact, Panel (b) below), is $1.9/1.4 = 1.36$, implying a pooled elasticity of $1/1.36 = 0.73$. This number is broadly in line with the elasticity calculated by Gabaix and Koijen (2022) using estimates from Lou (2010). However, as Section 3 argues, this calculation is a weighted average of several structural parameters and lacks clear economic interpretation for counterfactual analysis, motivating the structural estimation in Section 4.

Figure A.5: Portfolio Increases

(a) $\Delta \log S_{m,t+h}$ (b) $\Delta s^M_{i,t+h}$



Panel (a) of Figure A.5 presents impulse responses β_h estimated from

$$\Delta \log S_{m,t+h} = \alpha_m + \delta_t + \beta_h f_{m,t} + X_{i,t-1} + \varepsilon_{m,t+h} \text{ for } h = 0, 1, \ldots, 4,$$

where the dependent variable is value-weighted increase in mutual fund's current portfolio. α_m and δ_t are mutual fund and time fixed effects, and control variables $X_{i,t-1}$ include lagged portfolio weight change and flow shocks. β_h can be interpreted as the $h-$quarter ahead impulse response from the mutual fund flow shocks. Panel (b) presents the corresponding firm-level response, $\Delta s^M_{i,t+h}$, estimated from the baseline specification in equation (3). The sample is from 2007Q1 to 2021Q4.

A.3.2 Aggregate Flow Shocks

Figure A.6 plots the detrended aggregate flow shock δ_t together with aggregate profitability shock. The correlation between aggregate flow shock and aggregate profitability shock is 0.44. Intuitively, when fund clients are wealthier, they tend to delegate more money to mutual funds. However, aggregate profitability shock is not the sole driver of the flow shock fluctuations. For example, Dou et al. (2022) documents that heightened uncertainty drives aggregate outflow. [10]

Figure A.6: Aggregate Fund Flow Shocks and Aggregate Profitability Shocks

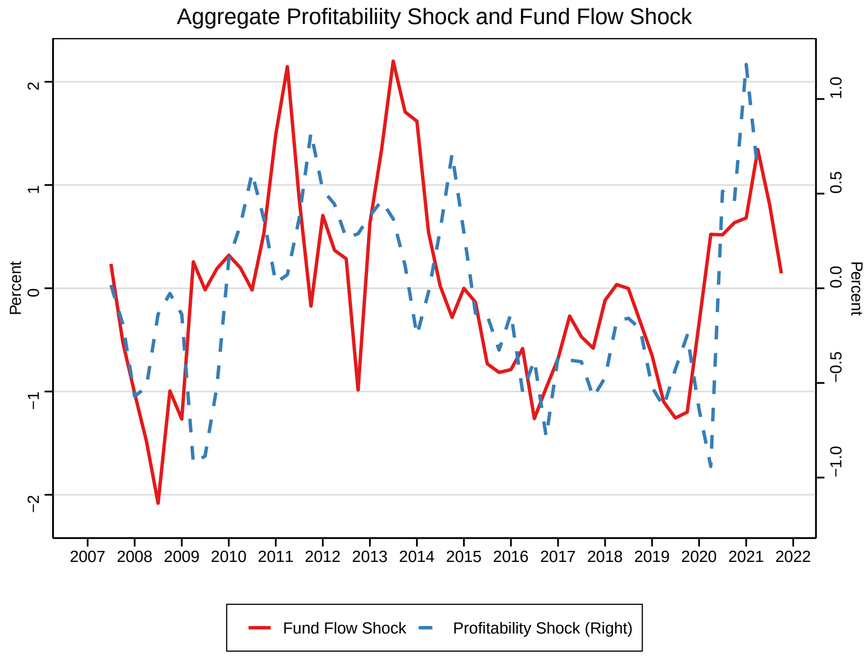


Figure A.6 presents the relationship between aggregate fund flow shock and aggregate profitability shock. Aggregate fund flow shocks are detrended time fixed effects δ_t in equation (1). To get aggregate profitability shocks, I first detrend profit per unit of real gross value added of nonfinancial corporate business from FRED, and estimate an AR(1) model extract the residuals. The correlation between the two shocks is 0.44, but profitability shocks are not the only driver of fund flow shocks.

[10] I do not explicitly model the delegation problem in this paper. Dou et al. (2022) provides a more comprehensive view on the drivers of aggregate flows.

B Model Appendix

B.1 Two-Period Model

B.1.1 $d \log \Lambda / d\phi > 0$

Plugging in the fund position $s^{M'}$ to (14), we get

$$\Lambda = \frac{1 + \frac{c_r \Omega \phi}{\beta I^\alpha}}{1 + c_r s_0^M}$$

$$\frac{d \log \Lambda}{d\phi} = \frac{c_r \Omega}{\beta I^\alpha (1 + c_r s_0^M) \Lambda}$$

$$= \frac{c_r \Omega}{\beta I^\alpha + c_r \Omega \phi} > 0$$

B.1.2 $d\Lambda / dI < 0$

Define $D = \gamma \sigma^2 + c_\phi$. Let $G(\phi, I)$ denote the first order condition of the mutual fund's problem (15), by the implicit function theorem,

$$\frac{d\phi}{dI} = -\frac{\partial G / \partial I}{\partial G / \partial \phi} = \frac{\alpha c_r \Omega \phi (2\beta I^\alpha + c_r \Omega \phi)}{I \left(D(\beta I^\alpha + c_r \Omega \phi)^2 + c_r \Omega (2\beta I^\alpha + c_r \Omega \phi) \right)}.$$

Using the definition of Λ in (14),

$$\frac{d\Lambda}{dI} = \frac{\partial \Lambda}{\partial I} + \frac{\partial \Lambda}{\partial \phi} \frac{d\phi}{dI}$$

$$= \Lambda \frac{-\alpha c_r \Omega \phi D (\beta I^\alpha + c_r \Omega \phi)}{I \left(D(\beta I^\alpha + c_r \Omega \phi)^2 + c_r \Omega (2\beta I^\alpha + c_r \Omega \phi) \right)} < 0$$

B.1.3 Proof of Model Properties

Proof of Property 2. Let $C = \mathbb{E} \log A - \log \beta - r_f + c_\phi \bar{\phi}$, Λ_I^ϕ denote the derivative of Λ with respect to I along the optimality path of ϕ, and $\lambda_I^\phi = \Lambda_I^\phi / \Lambda$. Using the two derivatives above, the two first order equations can be written as

$$F(\phi, I; \Omega) = \beta \alpha I^{\alpha - 1} - 1 + c_h \frac{(K_0^\alpha - I) + (K_0^\alpha - I)^2 \lambda_I^\phi (\phi, I, \Omega)}{K_0 \Lambda (\phi, I, \Omega)^2} = 0$$

$$G(\phi, I; \Omega) = D\phi + \frac{c_r \Omega \phi}{\beta I^\alpha + c_r \Omega \phi} - C + \log \left(1 + \frac{c_r \Omega \phi}{\beta I^\alpha} \right) - \log(1 + c_r s_0^M) = 0$$

where

$$\Lambda(\phi, I; \Omega) = \frac{1 + \frac{c_r \Omega \phi}{\beta I^\alpha}}{1 + c_r s_0^M}$$
$$\lambda_I^\phi(\phi, I; \Omega) = \frac{-\alpha c_r \Omega \phi D(\beta I^\alpha + c_r \Omega \phi)}{I\left(D(\beta I^\alpha + c_r \Omega \phi)^2 + c_r \Omega(2\beta I^\alpha + c_r \Omega \phi)\right)}.$$

Using the implicit function theorem,

$$\frac{dI}{d\Omega} = -\frac{F_\Omega G_\phi - F_\phi G_\Omega}{F_I G_\phi - F_\phi G_I},$$
$$\frac{d\phi}{d\Omega} = -\frac{F_I G_\Omega - F_\Omega G_I}{F_I G_\phi - F_\phi G_I},$$

It is straightforward to show that

$$F_I G_\phi - F_\phi G_I = \frac{\beta\alpha(\alpha-1)I^{\alpha-2}M}{(P+B)^2} - \frac{c_h}{K_0\Lambda^2}\frac{M}{(P+B)^2} + \frac{\alpha B D c_h}{K_0 \Lambda^2 I(P+B)}\left(4E + \frac{C_1}{I(P+B)M^2}E^2\right)$$
$$F_\Omega G_\phi - F_\phi G_\Omega = \frac{c_h B D}{K_0 \Lambda^2 (P+B)^2 \Omega}\left(-2(P+B)E + \frac{\alpha}{IM^2}C_2 E^2\right)$$
$$F_I G_\Omega - F_\Omega G_I = \frac{c_r \phi(2P+B)\beta\alpha(\alpha-1)I^{\alpha-2}}{(P+B)^2} + \frac{c_h}{K_0\Lambda^2(P+B)}\left(-\frac{c_r\phi(2P+B)}{(P+B)} + \frac{2c_r\alpha\phi D B}{IM}E + \frac{\alpha D B^2(2P+B)}{I^2 M \Omega}E^2\right)$$

where $B = c_r\Omega\phi$, $M = D(P+B)^2 + c_r\Omega(2P+B)$, $P = \beta I^\alpha$, $E = K_0^\alpha - I < 0$, and

$$C_1 = \big((1+\alpha)P + (1-2\alpha)B\big)M^2 + c_r\alpha\Omega(2B^2 + 3PB - 4P^2)M - c_r^2\alpha B\Omega^2(B+2P)(B+3P),$$
$$C_2 = D^2(P+B)^4(2B-P) + c_r\Omega\big[BD(3P+2B)(P+B)^2 + B\Omega(3P+2B)(2P+B) - c_r\Omega(2P+B)(B^2-2P^2)\big].$$

The E^2 terms in the Jacobians arise from differentiating the second-order issuance cost $c_h E^2 \lambda_I^\phi/(K_0\Lambda^2)$ in F and its cross-partial derivatives. Under the regularity condition that $|E\lambda_I^\phi|$ is sufficiently small—i.e., the price-pressure feedback on issuance costs is second-order relative to the direct issuance effect $c_h E/(K_0\Lambda^2)$—the leading terms dominate: $F_I G_\phi - F_\phi G_I < 0$, $F_\Omega G_\phi - F_\phi G_\Omega > 0$, $F_I G_\Omega - F_\Omega G_I < 0$. Therefore, $dI/d\Omega > 0$ and $d\phi/d\Omega < 0$. □

Proof of Property 1. First, note that

$$\frac{dP^*}{d\Omega} = \frac{\partial P^*}{\partial I}\frac{dI}{d\Omega} > 0$$

Second, from the definition of Λ^*,

$$\frac{d\Lambda^*}{d\Omega} = \frac{\partial \Lambda^*}{\partial \Omega} + \frac{\partial \Lambda^*}{\partial I}\frac{dI}{d\Omega} + \frac{\partial \Lambda^*}{\partial \phi}\frac{d\phi}{d\Omega}$$

From the definition $\Lambda = (P+B)/[P(1+c_r s_0^M)]$, the partial derivatives of Λ^* are

$$\frac{\partial \Lambda}{\partial \Omega} = \frac{B\Lambda}{\Omega(P+B)}, \qquad \frac{\partial \Lambda}{\partial I} = \frac{-\alpha B\Lambda}{I(P+B)}, \qquad \frac{\partial \Lambda}{\partial \phi} = \frac{B\Lambda}{\phi(P+B)}.$$

Substituting $dI/d\Omega$ and $d\phi/d\Omega$ from the implicit function theorem and factoring $B\Lambda/[\Omega(P+B)]$,

$$\frac{d\Lambda^*}{d\Omega} = \frac{B\Lambda D}{\Omega(P+B)} \cdot \frac{H + \dfrac{2c_h \alpha BE}{K_0\Lambda^2 I(P+B)}\left(1 - \dfrac{c_r\Omega}{M}\right)}{F_I G_\phi - F_\phi G_I} > 0,$$

where $H \equiv \beta\alpha(\alpha-1)I^{\alpha-2} - c_h/(K_0\Lambda^2)$. Both terms in the numerator are negative: $H < 0$, and $E < 0$ with $1 - c_r\Omega/M > 0$. The denominator $F_I G_\phi - F_\phi G_I < 0$ by the proof of Property 2. □

Proof of Property 3. The elasticities $ds^{R'}/dP$, $ds^{R'}/d\Lambda$ directly follow from the first order condition of residual investors' problem (13).

To derive the mutual fund's elasticities with respect to fundamental value and price pressure, we need to modify the system slightly so that there is an extra source of exogenous variation that shifts price pressure Λ but not from mutual fund flows. Consider a revised system that is summarized as

$$\begin{aligned} D\phi + \frac{d\log\Lambda}{d\phi}\phi &= C - \log\Lambda \\ \Lambda &= \Theta\frac{1 + \frac{c_r\Omega\phi}{\beta I^\alpha}}{1 + c_r s_0^M}, \end{aligned}$$

where Θ is an exogenous variable that shifts price pressure.

Note that $d\log\Lambda/d\phi$ is the same as before. The counterpart of mutual fund optimality $G(\cdot)$ can be rewritten as

$$D\phi + \frac{c_r\Omega}{P + c_r\Omega\phi}\phi = C - \log\Theta - \log\left(1 + \frac{c_r\Omega\phi}{P}\right) + \log(1 + c_r s_0^M)$$

Using the implicit function theorem, we get

$$\frac{d\phi}{d\log\Theta} = -\frac{1}{D + \frac{c_r\Omega}{P+c_r\Omega\phi} + \frac{c_r\Omega P}{(P+c_r\Omega\phi)^2}} < 0$$
$$\frac{d\phi}{d\log P} = \frac{c_r\Omega\phi(2P + c_r\Omega\phi)}{(P + c_r\Omega\phi)^2\left(D + \frac{c_r\Omega}{P+c_r\Omega\phi} + \frac{c_r\Omega P}{(P+c_r\Omega\phi)^2}\right)} > 0$$

□

B.2 Mutual Fund Utility

$$\begin{aligned}
&\frac{\beta}{1-\gamma_t}\mathbb{E}_t\left[e^{(1-\gamma_t)\tilde{q}_{i,t+1}}\right]\\
&= \frac{\beta}{1-\gamma_t}\mathbb{E}_t\left[e^{(1-\gamma_t)(q_{i,t}+r^M_{i,t+1})}\right]\\
&= \frac{\beta Q_t^{1-\gamma_t}}{1-\gamma_t}\mathbb{E}_t\left[e^{(1-\gamma_t)(r^M_{i,t+1})}\right]\\
&= \frac{\beta Q_t^{1-\gamma_t}}{1-\gamma_t}\mathbb{E}_t\left[\exp\left((1-\gamma_t)(r_f + \phi_{i,t+1}(r_{i,t+1} - r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma^2_{i,t})\right)\right]\\
&= \frac{\beta Q_t^{1-\gamma_t} R_f^{1-\gamma_t} 1}{1-\gamma_t}\mathbb{E}_t\left[\exp\left((1-\gamma_t)\left(\phi_{i,t+1}(r_{i,t+1} - r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma^2_{i,t}\right)\right)\right]\\
&= \frac{\beta Q_t^{1-\gamma_t} R_f^{1-\gamma_t}}{1-\gamma_t}\exp\left[(1-\gamma_t)\left(\phi_{i,t+1}(\mu_{i,t} - r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma^2_{i,t} + \frac{1}{2}(1-\gamma_t)\phi^2_{i,t+1}\sigma^2_{i,t}\right)\right]\\
&\equiv \frac{\beta Q_t^{1-\gamma_t} R_f^{1-\gamma_t}}{1-\gamma_t}\exp\left[(1-\gamma_t)u^M\right]
\end{aligned}$$

Therefore,

$$u^M = \frac{1}{1-\gamma_t}\log\mathbb{E}_t\left[R^{1-\gamma}\right] - r_f$$

Alternatively, if we assume $\tilde{q}_{i,t+1} = \bar{q} + r^M_{i,t+1} + \omega_{i,t+1}$, there will be an extra hedging motive to avoid firms that will be strongly affected by flow shocks.

B.3 Hedging Motives and Biased Holdings

Following Dou et al. (2022) and motivated by Figure A.6, to incorporate the aggregate component of fund flow shocks, suppose that the size of mutual fund i evolves following

$$\tilde{q}_{i,t+1} = q_{i,t} + r^M_{i,t+1} + \epsilon_{q,t+1},$$

where $\epsilon_{q,t+1}$ is the aggregate fund flow shock.

Using the same reshuffling assumption, the AUM available for investment is now time varying. Replace $\bar{q}$ with q_{t+1} in (33),

$$q_{i,t+1} = q_{t+1} + \omega_{i,t+1}, \tag{A.3}$$

where q_{t+1} is the average size of the mutual fund industry.

The aggregate average mutual fund size q_{t+1} and the aggregate profitability a_{t+1} follow a VAR(1) process:

$$\begin{bmatrix} q_{t+1} \\ a_{t+1} \end{bmatrix} = \begin{bmatrix} (1-\rho_q)\bar{q} \\ (1-\rho_a)\bar{a} \end{bmatrix} + \begin{bmatrix} \rho_q & 0 \\ 0 & \rho_a \end{bmatrix} \begin{bmatrix} q_t \\ a_t \end{bmatrix} + \begin{bmatrix} \epsilon_{q,t+1} \\ \epsilon_{a,t+1} \end{bmatrix}, \quad \begin{bmatrix} \epsilon_{q,t+1} \\ \epsilon_{a,t+1} \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} \sigma_q^2 & \rho_{q,a}\sigma_q\sigma_a \\ \rho_{q,a}\sigma_q\sigma_a & \sigma_a^2 \end{bmatrix}\right), \tag{A.4}$$

where $\bar{q}$ is the long-run mean of the log size of the mutual fund sector, ρ_q and ρ_a are the persistence parameters, and shocks $\epsilon_{q,t+1}$ and $\epsilon_{a,t+1}$ are jointly normal. σ_q and σ_a are their respective conditional volatility. $\rho_{q,a}$ is the correlation between the two shocks. Motivated by Figure A.6, $\rho_{q,a} > 0$ captures the occurrence of large inflow shocks during good times in a reduced-form manner.

Using the same Taylor approximation, tha manager's discounted payoff can be written as

$$\begin{aligned} u^M \approx & \frac{1}{1-\gamma_t}\mathbb{E}_t\left[\exp\left((1-\gamma_t)\left(\phi_{i,t+1}(r_{i,t+1}-r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma_{i,t}^2 + \epsilon_{q,t+1}\right)\right)\right] \\ = & \phi_{i,t+1}(\mu_{i,t}-r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma_{i,t}^2 + \frac{1}{2}(1-\gamma_t)\left(\phi_{i,t+1}^2\sigma_{i,t}^2 + 2\phi_{i,t+1}C_{i,t}\right), \end{aligned} \tag{A.5}$$

where $C_{i,t} = \mathrm{Cov}_t[r_{i,t+1}, \epsilon_{q,t+1}]$ is the conditional covariance between the return of firm i and aggregate flow shock $\epsilon_{q,t+1}$. The covariance term exists because positive aggregate profitability shocks drive up both firm production and returns, and these shocks typically coincide with positive mutual fund flow shocks ($\rho_{q,a} > 0$), Since $1-\gamma_t < 0$, higher covariance between returns and aggregate flow shocks reduces the fund manager's payoff. Therefore, fund managers are incentivized to underweight (overweight) firms that are more (less) cyclical to hedge against aggregate flow shocks.

Since ϵ_q and ϵ_a are jointly normal, we can write $\epsilon_q = \mathbb{E}[\epsilon_q|\epsilon_a] + \varepsilon_{qa} = \frac{\rho_{q,a}\sigma_q\sigma_a}{\sigma_a^2}\epsilon_a + \varepsilon_{qa}$, where $\text{Cov}(\epsilon_a, \varepsilon_{qa}) = 0$. Therefore, $C_i = \text{Cov}\left(\mathbb{E}[\epsilon_q|\epsilon_a], r_i\right) + \text{Cov}(\varepsilon_{qa}, r_i) = \tilde{C}_i + \text{Cov}(\varepsilon_{qa}, r_i)$. To capture the hedging incentive without materially changing the model structure, I use $\tilde{C}_{i,t} = \text{Cov}_t\left[r_{i,t+1}, \frac{\rho_{q,a}\sigma_q}{\sigma_a}\epsilon_{a,t+1}\right]$ to proxy for $C_{i,t}$. Since positive aggregate flow shocks likely raise stock prices, $\text{Cov}(\varepsilon_{qa}, r_i) > 0$, the proxy underestimates the hedging motives. To gauge the size of the error term, the variance of the residual is $\mathbb{V}(\varepsilon_{qa}) = \sigma_q^2(1 - \rho_{q,a}^2)$. Using the Cauchy-Schwarz inequality, $|\text{Cov}(\varepsilon_{qa}, r_i)| \leq \sqrt{\mathbb{V}(\varepsilon_{qa})\mathbb{V}(r_i)} = \sigma_r\sigma_q\sqrt{1 - \rho_{qa}^2}$.

B.4 Extension on Ownership Dynamics

It is natural to extend the model to include the changes in ownership and discuss its implication for misvaluation.

Given the issuance $H_{i,t}$ at market value $\tilde{P}_{i,t}$, current stockholders only hold $(\tilde{P}_{i,t} - H_{i,t})/\tilde{P}_{i,t}$ fraction of the firm. In other words, current stock holdings are shrunk to

$$s_{i,t}^{M,post} = \frac{\tilde{P}_{i,t} - H_{i,t}}{\tilde{P}_{i,t}} s_{i,t}^M \text{ and } s_{i,t}^{R,post} = \frac{\tilde{P}_{i,t} - H_{i,t}}{\tilde{P}_{i,t}} s_{i,t}^R \tag{A.6}$$

And the portion of new issuance equals $H_{i,t}/\tilde{P}_{i,t}$. Therefore, the new share holdings for residual investors should be written as

$$s_{i,t+1}^R = s_{i,t}^R \frac{\tilde{P}_{i,t} - H_{i,t}}{\tilde{P}_{i,t}} + \frac{V_{i,t} - \tilde{P}_{i,t}}{c_r \tilde{P}_{i,t}} \tag{A.7}$$

Note that we already calculated the targeted ownership share for investors, so we can calculate the change in their positions and calculate how costly it is to get them to purchase the new issuance, since newly issued shares are entirely purchased by the mutual funds and residual investors.

To get a simple analytical solution that is qualitatively similar, assume mutual fund adjustments are not affected by this effect. We can then calculate the incremental holding for residual investors:

$$s_{i,t+1}^R - s_{i,t}^R = \left(1 - \frac{\tilde{P}_{i,t} - H_{i,t}}{\tilde{P}_{i,t}}\right) s_{i,t}^R - \frac{V_{i,t} - \tilde{P}_{i,t}}{c_r \tilde{P}_{i,t}} = \frac{H_{i,t}}{\tilde{P}_{i,t}} s_{i,t}^R - \frac{V_{i,t} - \tilde{P}_{i,t}}{c_r \tilde{P}_{i,t}} \tag{A.8}$$

The more you issue, the less residual investors want to increase their holdings. So it is easier to get mutual funds to buy the new issuance.

In real life, it could be easier to get mutual funds to pay for the new issuance because they are bulk long-term investors whose postitions are relatively stable. Negotiating with the hedge funds and retailer investors (what residual investors really represent in this model) could be harder.

When we consider how shares are diluted, residual investor shares will shrink following (A.7).

Now that we have an additional term $h_{j,t} = H_{j,t}/P_{j,t}$ (and $\tilde{h}_{j,t} = H_{j,t}/\tilde{P}_{j,t}$), it becomes

$$\tilde{P}_{j,t} = \frac{1}{1 + c_h \left(s^M_{j,t} - s_{j,t+1} + \tilde{h}_{j,t} s^H_{j,t} \right)} P_{j,t} \tag{A.9}$$

$$= \frac{1 - c_h s^H_{j,t} h_{j,t}}{1 + c_h \left(s^M_{j,t} - s^M_{j,t+1} \right)} P_{j,t} \tag{A.10}$$

Under normal parameterization, when $H_{j,t}$ goes up, market value goes down. Note that this intuition is not entirely complete because of $s^M_{j,t+1}(\tilde{P}_{j,t})$ and $\tilde{h}_{j,t}(\tilde{P}_{j,t})$ are both functions of $\tilde{P}$.

B.5 First order conditions

The firm problem is

$$V_{i,t} = \max_{I_{i,t}, K_{i,t+1}, L_{i,t+1}} O_{i,t} + \mathbb{E}_t \left[M_{t,t+1} V_{i,t+1} \right],$$

where

$$\begin{aligned}
O_{i,t} &= \begin{cases} E_{i,t} - \Psi_{i,t} \equiv O^+(E_{i,t}, K_{i,t}, \Lambda_{i,t}(\phi_{i,t+1}(K_{i,t+1}, L_{i,t+1}))) & \text{if } E_{i,t} \le 0, \\ (1 - \tau_D)(E_{i,t} - B_{i,t}) + B_{i,t} - \Phi_{i,t} \equiv O^-(E_{i,t}, \Lambda_{i,t}(\phi_{i,t+1}(K_{i,t+1}, L_{i,t+1}))) & \text{if } E_{i,t} > 0. \end{cases} \\
E_{i,t} &= (1 - \tau) A_t^{b_i} X_{i,t} K^\alpha_{i,t} + \tau \delta K_{i,t} + \tau r_f L_{i,t} \mathbb{1}_{\{L_{i,t} > 0\}} - I_{i,t} - G_{i,t} + L_{i,t+1} - (1 + r_l) L_{i,t} \\
K_{i,t+1} &= (1 - \delta) K_{i,t} + I_{i,t} \\
G_{i,t} &= \frac{c_k}{2} \left(\frac{I_{i,t}}{K_{i,t}} \right)^2 K_{i,t} \\
\Psi_{i,t} &= c_{h0} \frac{K_{i,t}}{\Lambda_{i,t}} + \frac{c_{h1}}{2} \left(\frac{H_{i,t}}{\Lambda_{i,t} K_{i,t}} \right)^2 K_{i,t} \\
\Phi_{i,t} &= \frac{c_b}{2} \left(\frac{\Lambda_{i,t} B_{i,t}}{E_{i,t}} \right)^2 E_{i,t} \\
L_{i,t+1} &\le \varphi K_{i,t+1}.
\end{aligned}$$

First, when $E_{i,t} > 0$, $B_{i,t} = \frac{\tau_D}{c_b \Lambda^2} E_{i,t}$ and $O_{i,t} \equiv O^+_{i,t} = \left(1 - \tau_D + \frac{1}{2} \frac{\tau_D^2}{c_b \Lambda^2} \right) E_{i,t}$. Let q_t and λ_t denote the Lagrangian multipliers associated with (20) and (22). The first-order conditions with respect to I_t, K_{t+1}, and L_{t+1} are

$$-\frac{\partial O}{\partial I} = q \tag{A.11}$$

$$q = \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial K'}\right]\right\} + \lambda\varphi + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial K'} \tag{A.12}$$

$$\lambda - \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial L'}\right]\right\} = \frac{\partial O}{\partial E}\frac{\partial E}{\partial L'} + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial L'} \tag{A.13}$$

C Computation Appendix

C.1 Algorithm to Solve the Model

1. Guess portfolio holding rule $\phi'^0(K', L', S)$, value function $V^0(S)$, and investment decision $K'^0(S)$.

2. Market clearing implies $s^{R'}(\phi'^0(K', L', S), K', L', S) = s^{R'0}(K', L', S)$ and therefore we have $\Xi(s^{R'0}(K', L', S), K', L', S) = \Xi^0(K', L', S)$. We also have the $\Xi^0(K'^0(S), L'^0(S), S) = \Xi^0(S)$.

3. Use $\Xi^0(K', L', S)$, run the Value Function Iteration routine. Get $V^1(S)$ and policy functions $K'^1(S)$.

4. Given $\Xi^0(S)$ (for tomorrow's value) and $V^0(S)$, calculate $\phi'^1(K', S)$.

5. Compare distances $|\phi^1(K', S) - \phi^0(K', S)|$ and $|V^1(S) - V^0(S)|$. Update $\phi^0(K', S)$ and $V^0(S)$. If larger than tolerance, go back to step 2.

C.2 Algorithm for the Indirect Inference Estimation

C.2.1 Constructing Model Moments and the Weight Matrix

I first regress firm returns, NEI, mutual fund portfolio holding, mutual fund positions, and changes in mutual fund positions on time and firm fixed effects and extract the residual to demean all relevant variables by firm and time. I calculate two dummies indicating non-negative and non-positive NEI (excluding dividends), and use the non-demeaned NEI data to calculate the conditional means. After calculating one-period forward of returns, I drop observations with any missing variable. This results in a panel of $N = 156,080$ observations. To keep the consistency between data and the model, I calculate the impulse response functions without added firm controls. As shown in Table 6, the regression coefficients are qualitatively similar to the baseline specification reported in Figure 1.

I follow Erickson and Whited (2002) to calculate the influence functions and covary the influence functions to calculate the weight matrix.

C.2.2 Estimation

Let x_n be a vector of data of dimension J, where J is the number of relevant variables. Let b be the vector of structural parameters to be estimated. Let $y_{n,k}(b)$ be the J-dimensional simulated data vector from simulation $k = 1, \ldots, K$. Let $x = \{x_1, \ldots, x_N\}$ and $y_k(b) = \{y_{1,k}, \ldots, y_{N,k}\}$ denote the data panel and the k-th simulated panel. I estimate b by matchign a set of simulated moments, denoted as $h(y_k(b))$, with the corresponding set of actual data moments, denoted as $h(x)$. Define

$$g(x,b) = h(x) - \frac{1}{K}\sum_{k=1}^{K} h(y_k(b)). \tag{A.14}$$

The indirect inference estimator of b is then defined as the solution to the minimization of

$$\hat{b} = \arg\min_{b} g(x,b)'\hat{W} g(x,b), \tag{A.15}$$

where $\hat{W}$ is the weight matrix calculated from the influence functions.

C.3 Robustness: Heterogeneous Productivity Exposure

The counterfactual results in Table 11 set firms' aggregate-shock exposure b_i in the production function $Y_{i,t} = A_t^{b_i} X_{i,t} K_{i,t}^{\alpha}$ to $b_i = 1$ for all firms. As a robustness check, I re-solve and re-simulate the model letting $b_i \sim \mathcal{N}(1, 0.2)$ vary across firms, following David et al. (2022) and Choi et al. (2021), and repeat the same four counterfactual scenarios. Table A.3 reports the results.

Table A.3: Capital Misallocation with Heterogeneous Productivity Exposure: Percentage Change from Baseline

Param	Base	CF	Description	Agg TFP	SD(NEI/K)	SD(I/K)	SD(R)
$\sigma_{\omega i}$	0.210	0.240	High flow vol.	-0.007	0.420	-0.216	0.881
$\exp(\bar{q})$	0.290	0.220	Small fund sector	0.018	0.059	-0.594	0.678
c_h	0.878	1.000	High capacity constr.	-0.011	-0.320	0.165	-0.643
c_k	0.492	0.450	Low capital adj. cost	0.150	0.953	5.277	0.276

Table A.3 repeats Table 11's four counterfactual scenarios with heterogeneous firm exposure to aggregate productivity shocks, $b_i \sim \mathcal{N}(1, 0.2)$, in place of the baseline's $b_i = 1$. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded.

The results are robust to this alternative specification. Reducing the capital adjustment cost still yields by far the largest aggregate TFP effect (0.15%), while the three financial-side counterfactuals remain an order of magnitude smaller, with the same signs as in the baseline: higher flow shock volatility lowers TFP (-0.007%), a smaller captive mutual fund sector raises TFP (0.018%), and less elastic residual investors lower TFP (-0.011%). This confirms that the paper's central finding, that financial-side frictions generate only modest capital misallocation relative to real investment frictions, does not depend on assuming firms are ex-ante homogeneous in their exposure to aggregate productivity shocks.